Supplementary Material — The Nostalgic Undertow

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1. S1. Proof of Theorem 1 (the adiabatic identity $B = K/2$ and its transient realisability, OU)

Theorem 1 main states that in the adiabatic OU limit the cumulative excess loss grows as $L_{\text{excess}}(t) = At + (K/2)\ln(\lambda t) + C + o(1)$ with logarithmic coefficient equal to $K/2$, **on the transient concentration window** $t \lesssim \tau_{\text{conc}}$ (the duration of the prior-collapse burst, S1.3a–c); asymptotically, for $t \gg \tau_{\text{conc}}$, the steady-state gain $g^\star$ saturates the accumulation and the logarithm passes into a constant plateau $\sim (K/2)\ln(1/g^\star)$ (S1.8a) — a structural transience analysed in S1.4–S1.5. The derivation is carried out in the linear-Gaussian (Kalman) approximation of the observational model and under the explicit assumptions (A1)–(A3) of the statement of Theorem 1 main; in particular, step (S1.4) relies on the heuristic reduction $n_{\text{eff}} \asymp \lambda s$ to class II (A3) *on the transient window*, while the rigorous transient non-stationary filter is left open (§ 7 main). Within this framework the coefficient $K/2$ depends neither on σ nor on the curvature of the model. The derivation proceeds in five steps: (S1.1) reduction to a scalar tracking problem over independent coordinates; (S1.2) the tracking error of the EWMA filter in an OU environment, the Riccati solution, the steady-state $g^\star$, and the two time scales (the forgetting scale τ_f^{forget} and the concentration scale τ_{conc}); (S1.3) expansion of the instantaneous excess loss into a quadratic form in the Fisher metric; (S1.4) integration over the transient window, derivation of the coefficient $K/2$, and the passage of the logarithm into a plateau for $t \gg \tau_{\text{conc}}$; (S1.5) analysis of the finite-amplitude correction $g(\epsilon)$ and the structural unattainability of $\hat{B} \to K/2$ as a slope; (S1.6) optimality of the floor as a lower bound over all memory architectures (sufficiency of the Kalman posterior + MMSE, in the linear-Gaussian window A1).

1.1. S1.1 Tensorisation over independent coordinates

By construction of the OU parametrization (2.1) main, the increments dW_{ij} are independent over (i, j), the covariance is diagonal, and the process $\theta(t) = (\theta_{ij}(t))$ decomposes into $K = k(k-1)$ independent scalar OU processes

$$d\theta_{ij} = -\lambda(\theta_{ij} - \theta^*_{ij})\, dt + \sigma\, dW_{ij}, \qquad \mathbb{E}[\theta_{ij}(t)\theta_{ij}(s)] = \sigma_0^2\, e^{-\lambda|t-s|}.$$

For the ideal Bayesian learner the posterior over $\theta(t)$ factorises over the coordinates at leading order in the adiabatic limit (per-bit uniformity, [1] Remark 5, § S2); consequently both the instantaneous excess loss and its cumulative sum are additive:

$$L_{\text{excess}}(t) = \sum_{ij} L^{(ij)}_{\text{excess}}(t) \cdot (1 + o(1)), \tag{S1.1}$$

where $L^{(ij)}_{\text{excess}}$ is the excess loss on a single coordinate. The additivity of mutual information and of the KL divergence over independent coordinates is a standard identity [2] ch. 2.5; unlike the χ^2 measure (multiplicative under independence), KL/MI are additive, which is precisely what produces the factor K in the final coefficient (the same logic as in [1] § S1.5). It suffices to prove the scalar result $L^{(ij)}_{\text{excess}}(t) = at + \frac{1}{2}\ln(\lambda t) + c + o(1)$ and sum over the K coordinates.

Coordinate-free derivation of the coefficient $K/2$ (the load-bearing argument). The componentwise tensorisation (S1.1) is convenient as an illustration, but the *value* of the logarithmic coefficient $K/2$ does not depend on it and does not rest on the diagonality of the Fisher information or the independence of the coordinates. In matrix form, the logarithmic term of the transient concentration (S1.4–S1.8 below) is the trace of the product of the Fisher metric and the transient posterior covariance: under concentration of the posterior with effective number of observations n_{eff}, the transient covariance over θ is $\Sigma_{\text{tr}} = (\mathcal{I}\, n_{\text{eff}})^{-1}$ (Bayesian concentration for a K-dimensional parameter), and the instantaneous logarithmic contribution to the excess loss is a quadratic form (S1.5) with trace

$$\tfrac{1}{2}\operatorname{tr}\big(\mathcal{I}\cdot\Sigma_{\text{tr}}\big) = \tfrac{1}{2}\operatorname{tr}\big(\mathcal{I}\,(\mathcal{I}\,n_{\text{eff}})^{-1}\big) = \tfrac{1}{2}\operatorname{tr}(\mathrm{Id}_K)/n_{\text{eff}} = \frac{K}{2\,n_{\text{eff}}},$$

since $\mathcal{I}\,(\mathcal{I})^{-1} = \mathrm{Id}_K$ for *any* SPD Fisher matrix $\mathcal{I}$ (diagonal or not), and $\operatorname{tr}(\mathrm{Id}_K) = K$. The Fisher information cancels *identically* — inside the trace, without appeal to a coordinatewise decomposition — so the coefficient $K/2$ holds even when the softmax couples the logits of one row (Fisher non-diagonal): it depends only on the dimension $K = \dim\theta$, not on the structure of $\mathcal{I}$ and not on per-bit uniformity. Substituting $n_{\text{eff}} \asymp \lambda s$ (S1.4) and integrating $\int (K/2)\, ds/s$ yields (S1.10) directly. Per-bit uniformity remains needed only for the *floor* C2 (§ S2, additivity of the contribution of fresh bits), but *not* for the value of $K/2$. The coordinatewise derivation (S1.1)–(S1.10) below is the special (diagonal) case of this trace, given here for concreteness.

1.2. S1.2 Tracking error of the EWMA filter and the Riccati equation

Consider a scalar coordinate $\theta(t)$ (the index ij omitted) with hidden OU dynamics $d\theta = -\lambda\theta\, dt + \sigma\, dW$ (anchor $\theta^* = 0$ without loss of generality). The observations are discrete (one measurement per step $\Delta t = 1$, as in the simulations). In the linear-Gaussian approximation around the current estimate, an observation is equivalent to a noisy measurement $dy_s = \theta(s)\, ds + \mathcal{I}_{\text{rate}}^{-1/2}\, dV_s$, where V_s is a standard Wiener process and $\mathcal{I}_{\text{rate}}$ is the *rate of accumulation of Fisher information* (Fisher rate), of dimension [Fisher]·time^{-1}: the amount of Fisher information about θ arriving per unit time. Under discretisation $\Delta t = 1$ with one observation per step, $\mathcal{I}_{\text{rate}} = \mathcal{I}$ numerically (the information per step equals the information per unit time), and below we write $\mathcal{I}$ in the sense of $\mathcal{I}_{\text{rate}}$; in the general case the correspondence is $\mathcal{I}_{\text{rate}} = \mathcal{I}/\Delta t$.

The optimal Bayesian filter for a linear-Gaussian system with hidden OU state is the stationary Kalman–Bucy filter [3, 4]. The stationary posterior variance (tracking error) $\Sigma_\infty := \lim_{t\to\infty} \mathbb{E}[(\hat{\theta}(t) - \theta(t))^2]$ is the positive root of the continuous-time algebraic Riccati equation (with Fisher rate $\mathcal{I}_{\text{rate}}$, dimensions consistent: $[\lambda\Sigma] = [\sigma^2] = [\mathcal{I}_{\text{rate}}\Sigma^2]$ — variance·time^{-1})

$$-2\lambda\Sigma_\infty + \sigma^2 - \mathcal{I}_{\text{rate}}\,\Sigma_\infty^2 = 0 \implies \Sigma_\infty = \frac{-2\lambda + \sqrt{4\lambda^2 + 4\mathcal{I}_{\text{rate}}\sigma^2}}{2\mathcal{I}_{\text{rate}}} = \frac{\lambda}{\mathcal{I}_{\text{rate}}}\left(\sqrt{1 + \frac{\mathcal{I}_{\text{rate}}\sigma^2}{\lambda^2}} - 1\right). \tag{S1.2}$$

The factor 2 in the term $-2\lambda\Sigma_\infty$ is standard for OU (the generator $-\lambda$ acts on the variance with a doubling); the controlling dimensionless parameter of *tracking* convergence is exactly $\epsilon_{\text{track}} := \mathcal{I}_{\text{rate}}\sigma^2/\lambda^2$ (definition (2.2) main), that is, the ratio of the Fisher rate to the drift rate, weighted by the noise amplitude. We emphasise the distinction from $\epsilon = \sigma^2/\lambda$: ϵ is the concentration of the OU latent itself, whereas $\epsilon_{\text{track}} = \mathcal{I}_{\text{rate}}\epsilon/\lambda$ is the concentration of the filter's *estimate* of the latent; it is precisely ϵ_{track}, not ϵ, that controls the expansion below. In the adiabatic limit $\epsilon_{\text{track}} \ll 1$, expanding the square root gives

$$\Sigma_\infty = \frac{\lambda}{\mathcal{I}} \cdot \frac{\mathcal{I}\sigma^2}{2\lambda^2}\left(1 + O\!\left(\tfrac{\mathcal{I}\sigma^2}{\lambda^2}\right)\right) = \frac{\sigma^2}{2\lambda}\,(1 + O(\epsilon_{\text{track}})) = \sigma_0^2\,(1 + O(\epsilon_{\text{track}})). \tag{S1.3}$$

This is the *stationary* residual filter lag: $\Sigma_\infty = \sigma_0^2(1 + O(\epsilon_{\text{track}})) \propto \sigma^2/\lambda \to 0$ in the OU-concentration limit. The equivalent *forgetting scale* of the filter is $\tau_f^{\text{forget}} = 1/\sqrt{\lambda^2 + \mathcal{I}\sigma^2} \asymp 1/\lambda = \tau_E$ in the adiabatic limit — that is, the optimal filter forgets on the scale of the environmental drift. *(Terminological clarification: two scales are distinguished below. This one — the forgetting scale $\tau_f^{forget} \asymp \tau_E$ — sets the memory depth of the filter. A separate scale τ_{conc} (the time to reach the steady-state $g^\star$, S1.3a) sets the duration of the transient concentration burst; in the adiabatic limit they diverge as $\tau_{conc}/\tau_f^{forget} \asymp 1/g^\star = n_{eff}^\star \to \infty$ (and not $\asymp 1/\epsilon_{track}$: since $g^\star = \frac{1}{2}\epsilon_{track}\lambda$, we have $1/g^\star = 2/(\epsilon_{track}\lambda)$, see below).)*

Steady-state gain and the concentration floor. The stationary Kalman filter has a steady-state gain $g^\star := \mathcal{I}_{\text{rate}}\Sigma_\infty/(1 + \mathcal{I}_{\text{rate}}\Sigma_\infty\,\Delta t)$, in the adiabatic limit at $\Delta t = 1$ with leading term

$$g^\star = \mathcal{I}_{\text{rate}}\,\Sigma_\infty\,(1 + O(\epsilon_{\text{track}})) = \mathcal{I}_{\text{rate}} \cdot \frac{\sigma^2}{2\lambda}\,(1 + O(\epsilon_{\text{track}})) = \tfrac{1}{2}\,\epsilon_{\text{track}}\,\lambda\,(1 + O(\epsilon_{\text{track}})) \;\propto\; \sigma^2, \tag{S1.3a}$$

that is, $g^\star \to 0$ in the adiabatic limit, with $g^\star \propto \sigma^2$ (linear in the forcing variance at fixed $\lambda, \mathcal{I}_{\text{rate}}$). This $g^\star$ sets the *floor of the effective number of observations*: the steady-state filter weights the past geometrically with factor $(1 - g^\star)$ per step, so the effective number of accumulated informative observations saturates at

$$n_{\text{eff}}^\star = \frac{1 - g^\star}{g^\star} \approx \frac{1}{g^\star} \quad (\text{exactly } (1 - g^\star)/g^\star,\ \text{leading term } 1/g^\star), \tag{S1.3b}$$

rather than growing without bound. The characteristic time to reach this floor is the *concentration time*

$$\tau_{\rm conc} \asymp n^\star_{\rm eff} \asymp 1/g^\star \;\propto\; 1/\sigma^2, \tag{S1.3c}$$

the number of steps over which the geometric window of the filter fills up. Crucially: $\tau_{\rm conc} \asymp 1/g^\star \propto 1/\sigma^2$ is a *different* scale from the forgetting scale $\tau_f^{\rm forget} \asymp \tau_E = 1/\lambda$; in the adiabatic limit their ratio is $\tau_{\rm conc}/\tau_E \asymp 1/g^\star = n^\star_{\rm eff} \to \infty$ (the physical $\tau_{\rm conc} = 1/(\lambda g^\star)$, divided by $\tau_E = 1/\lambda$, gives exactly $1/g^\star = n^\star_{\rm eff}$ — the number of effective observations). Since $g^\star = \frac{1}{2}\epsilon_{\rm track}\lambda$ (S1.3a), this is $1/g^\star = 2/(\epsilon_{\rm track}\lambda)$ — it contains the explicit factor $1/\lambda$, so the ratio $\tau_{\rm conc}/\tau_E$ is $1/g^\star = n^\star_{\rm eff}$, and *not* $1/\epsilon_{\rm track}$. It is precisely $\tau_{\rm conc}$, not τ_E, that bounds the window of logarithmic growth (S1.4), and its upper bound in adiabatic time is $\lambda\tau_{\rm conc} \asymp 1/g^\star = n^\star_{\rm eff}$.

Physical origin of the adiabatic bound $1/\sigma^2$. The upper bound of the adiabatic window $t \ll 1/\sigma^2$ has a direct physical meaning: $1/\sigma^2$ is the characteristic time of an $O(1)$ diffusive excursion of the OU latent. Over a time t the accumulated variance of the purely diffusive part of the OU increment is $\sigma^2 t$; at $t \sim 1/\sigma^2$ it reaches $O(1)$, the latent has had time to shift by a scale comparable to the variable itself, and the softmax linearisation (A1) around the current estimate loses validity (the linearisation point drifts out of the neighbourhood in which the remainder of the EKF/Laplace approximation is controllable). Therefore $t \ll 1/\sigma^2$ is not a technical but a physical limitation on the domain of validity of the linear-Gaussian approximation.

Why the window is not opened by the adiabatic limit (exact statement of inseparability). The earlier formulation — "$\tau_{\rm conc}$ and the ceiling $1/\sigma^2$ are both $\propto 1/\sigma^2$, hence not separable" — is *physically incomplete*: the mere coincidence of the powers of σ does not entail inseparability, since two quantities, both $\propto 1/\sigma^2$, may have any constant ratio. The correct argument goes through the *width of the log-window* and its σ-invariance. The available log-carrying window is $[t_{\min}, t_{\max}]$, with lower bound $t_{\min} \asymp$ a few $\tau_{\rm conc} \asymp 1/g^\star$ (the logarithm is alive only after the concentration burst has subsided) and upper bound $t_{\max} \asymp c_0/\sigma^2$ (the adiabatic ceiling). Its *logarithmic width* (ratio of bounds) is

$$W = \frac{t_{\max}}{t_{\min}} \asymp \frac{c_0/\sigma^2}{1/g^\star} = c_0\,\frac{g^\star}{\sigma^2} \asymp \frac{\mathcal{I}_{\rm rate}}{\lambda} = \frac{1}{\lambda r}$$

(having substituted $g^\star = \frac{1}{2}\mathcal{I}_{\rm rate}\sigma^2/\lambda$ from (S1.3a), σ^2 *cancels*; $r = 1/\mathcal{I}_{\rm rate}$ is the observation noise). The decisive point: **the window width $W \asymp 1/(\lambda r)$ is σ-invariant** — σ has dropped out. Therefore the adiabatic limit $\sigma \to 0$ ($\epsilon_{\rm track} \to 0$) *does not open* the log-window: both the lower bound $\tau_{\rm conc} \propto 1/\sigma^2$ and the upper bound $1/\sigma^2$ stretch *in the same proportion*, the ratio remaining fixed at $\asymp 1/(\lambda r)$. This is the genuine cause of inseparability, and it *refutes* the working hypothesis ("push $\epsilon_{\rm track} \to 0$ and the window will open"): reducing σ does not touch W. The window W can be widened only by increasing the Fisher rate per correlation interval of the environment, $\mathcal{I}_{\rm rate}/\lambda = 1/(\lambda r) \to \infty$ (infinitely precise observation $r \to 0$, or $\lambda \to 0$ faster than $\mathcal{I}_{\rm rate}$). But this *takes one out of the slow-drift (adiabatic) regime*: $\mathcal{I}_{\rm rate}/\lambda \to \infty$ means that $\epsilon_{\rm track} = \mathcal{I}_{\rm rate}\sigma^2/\lambda^2 = (\mathcal{I}_{\rm rate}/\lambda)(\sigma^2/\lambda)$ ceases to be small at fixed concentration of the latent, violating the slow-driving condition $\lambda\tau_{\mathrm{relax}} \ll 1$ and the quasi-stationarity of θ on which (A1) and the entire derivation stand. Conclusion: the inseparability of the log-carrying and asymptotic windows is a *consequence of the adiabatic constraint* (fixed Fisher rate per correlation interval), not of an arithmetic coincidence of the powers of σ; in any regime where (A1) and quasi-stationarity hold, $W \asymp 1/(\lambda r) = O(1)$ relative to σ, and the log-window does not open. This is the key to the structural unattainability of S1.5.

In addition to the stationary part, the filter has a *transient* component: starting from prior uncertainty, the posterior variance $\Sigma(t)$ decreases toward Σ_∞. For slow drift, the relevant transient variance is controlled by the accumulation of *informative* observations. **In the transient concentration regime** ($s \lesssim \tau_{\mathrm{conc}}$, before reaching the steady-state $g^\star$), observations separated by more than τ_E decorrelate (the covariance of θ decays as $e^{-\lambda|t-s|}$), and the effective number of independent observations about the current $\theta(t)$ grows as

$$n_{\mathrm{eff}}(s) \asymp \frac{s}{\tau_E} = \lambda s \qquad \text{(transiently, } s \lesssim \tau_{\mathrm{conc}}\text{).} \tag{S1.4}$$

This is the *transient* (burst) regime: the growth (S1.4) does not continue up to $t \to \infty$ — for $s \gtrsim \tau_{\mathrm{conc}}$ the steady-state $g^\star$ (S1.3a) cuts off the accumulation, and n_{eff} saturates at the floor $n^\star_{\mathrm{eff}} = (1-g^\star)/g^\star \approx 1/g^\star$ (S1.3b). The linear growth λs and the saturation at $1/g^\star$ coincide at $s \asymp \tau_{\mathrm{conc}}$, which fixes the transition point. *Clarification of time units (to avoid an apparent double definition).* The concentration time is naturally written in two consistent normalisations: (a) in *physical* time (number of steps s, $\Delta t = 1$) the transition $n_{\mathrm{eff}}(s) = \lambda s = n^\star_{\mathrm{eff}} = 1/g^\star$ gives $\tau^{\mathrm{phys}}_{\mathrm{conc}} \asymp 1/(\lambda g^\star)$ — it is precisely this quantity that is measured by the number of steps until the filter reaches the plateau; (b) in *adiabatic* time $u = \lambda s$ (number of correlation intervals of the environment) the same transition is $u_{\mathrm{conc}} = \lambda\, \tau^{\mathrm{phys}}_{\mathrm{conc}} \asymp 1/g^\star$ — this is the form (S1.3c), written in dimensionless u (equal to $n^\star_{\mathrm{eff}}$, the number of effective observations). The relation is $\tau^{\mathrm{phys}}_{\mathrm{conc}} = u_{\mathrm{conc}}/\lambda$, and both are $\propto 1/\sigma^2$ (since $g^\star \propto \sigma^2$). Below, τ_{conc} without a marker denotes the physical scale $1/(\lambda g^\star)$, except in (S1.3c), where it is taken in adiabatic u. Crucially: the inseparability of τ_{conc} and the adiabatic ceiling $1/\sigma^2$ (the key to S1.5) holds in the *consistent* normalisation — in physical time both are $\asymp 1/\sigma^2$, in adiabatic time both are $\asymp \lambda/\sigma^2$; the ratio on which the structural unattainability stands is not changed by the normalisation. In the transient window the posterior variance about the "current level" of the latent decreases as $\Sigma_{\mathrm{tr}}(s) \asymp 1/(\mathcal{I}\, n_{\mathrm{eff}}(s)) = 1/(\mathcal{I}\lambda s)$ — standard Bayesian concentration $1/(\mathcal{I}n)$ with effective $n = \lambda s$ [5]; asymptotically $\Sigma_{\mathrm{tr}} \to 1/(\mathcal{I}n^\star_{\mathrm{eff}}) = \Sigma_\infty$ (the floor merges with the stationary lag).

1.3. S1.3 Quadratic expansion of the instantaneous excess loss

The instantaneous increment of the excess loss at step s is the log-likelihood ratio of the oracle and the learner:

$$\Delta L_{\text{excess}}(s) = \ln \frac{p(X_s \mid \theta(s))}{\hat{p}(X_s \mid \hat{\theta}(s))}.$$

Averaging over $X_s \sim p(\cdot \mid \theta(s))$ gives the KL divergence $D_{KL}(p(\cdot \mid \theta(s)) \,\|\, \hat{p}(\cdot \mid \hat{\theta}(s)))$. For a regular parametric model, at small tracking error $\delta\theta := \hat{\theta}(s) - \theta(s)$ the Taylor expansion of the KL to second order (the first order vanishes at the point $\theta(s)$) reads

$$\mathbb{E}[\Delta L_{\text{excess}}(s)] = \tfrac{1}{2}\,\mathcal{I}(\theta(s))\,\mathbb{E}[\delta\theta^2] + O(\mathbb{E}|\delta\theta|^3) \tag{S1.5}$$

— the standard quadratic expansion of the regret in the Fisher metric [5, 6, 2] ch. 11. The tracking error $\mathbb{E}[\delta\theta^2]$ decomposes into a stationary and a transient part (S1.2 and S1.4):

$$\mathbb{E}[\delta\theta^2(s)] = \underbrace{\Sigma_\infty}_{\text{stationary}} + \underbrace{\Sigma_{\text{tr}}(s)}_{\text{transient}} = \sigma_0^2(1 + O(\epsilon_{\text{track}})) + \frac{1}{\mathcal{I}\lambda s}(1 + o(1)). \tag{S1.6}$$

1.4. S1.4 Integration and derivation of the coefficient $K/2$

Substituting (S1.6) into (S1.5) and integrating over s from some s_0 (after the initial transient) **on the transient concentration window** $s_0 \leq s \lesssim \tau_{\mathbf{conc}}$, where the transient variance is still decreasing by (S1.4):

$$L^{(ij)}_{\text{excess}}(t) = \int_{s_0}^{t} \tfrac{1}{2}\mathcal{I}\left[\Sigma_\infty + \Sigma_{\text{tr}}(s)\right] ds + \text{const} = \underbrace{\tfrac{1}{2}\mathcal{I}\Sigma_\infty\,(t - s_0)}_{\text{linear } a\,t} + \underbrace{\frac{1}{2}\int_{s_0}^{\min(t,\,\tau_{\text{conc}})} \frac{ds}{s}}_{\text{logarithmic term (S1.8)}}. \tag{S1.7}$$

The upper limit of the logarithmic integral is $\min(t, \tau_{\mathbf{conc}})$, **not** t: the integrand $\Sigma_{\text{tr}}(s) \asymp 1/(\mathcal{I}\lambda s)$ holds only while $n_{\text{eff}}(s) \asymp \lambda s$ grows (S1.4), that is, on the transient window $s \lesssim \tau_{\text{conc}}$. For $s \gtrsim \tau_{\text{conc}}$ the steady-state $g^\star$ saturates n_{eff} at the floor $1/g^\star$, the transient variance reaches Σ_∞, and the integrand Σ_{tr} ceases to give a $1/s$ contribution — it is absorbed into the stationary term Σ_∞. This is precisely the structural cause of the transience of the logarithm, built into the mechanism rather than an external caveat.

More precisely, the transient contribution to (S1.5) is $\frac{1}{2}\mathcal{I}\cdot\Sigma_{\text{tr}}(s) = \frac{1}{2}\mathcal{I}\cdot\frac{1}{\mathcal{I}\lambda s} = \frac{1}{2\lambda s}$ per unit of *physical* time (on the transient window). The reduction to *adiabatic* time $u = \lambda s$ (number of correlation intervals) is a substantive step relying on assumption (A3) about class-II concentration: the passage to $n_{\text{eff}} \asymp \lambda s$ is *heuristically motivated* by analogy with Bartlett's effective sample size $n_{\text{eff}} = n/(1 + 2\sum_s \rho(s))$ [7] for observations with exponential autocorrelation $\rho(s) = e^{-\lambda|s|}$. *Carefully on the factor* $\frac{1}{2}$*:* for $\rho(s) = e^{-\lambda|s|}$ the sum of autocorrelations is $1 + 2\sum_{s\geq 1}\rho(s) \approx 2\sum_{s\geq 0} e^{-\lambda s} \approx 2/\lambda$ at $\lambda \ll 1$, whence $n_{\text{eff}} = n/(1 + 2\sum_s \rho) \approx n/(2/\lambda) = (\lambda/2)\,n$ — that is, with a factor $\frac{1}{2}$, $n_{\text{eff}} \approx (\lambda/2)\,n$, and *not* λn. This factor $\frac{1}{2}$ does not vanish without trace: it enters only the *additive logarithmic constant*, not the slope, since $\ln\big((\lambda/2)s\big) = \ln(\lambda s) - \ln 2$ — the shift $-\ln 2$ per coordinate is absorbed into the constant c_{ij} (and further into C) of formula (S1.9)–(S1.10), while the coefficient of $\ln(\lambda s)$ remains $\frac{1}{2}$. We therefore write $n_{\text{eff}} \asymp \lambda n$ *as an order-exact relation up to a multiplicative constant* $O(1)$ (here $\frac{1}{2}$), fixing only the *exponent* λs and hence the slope $K/2$; the exact value of the constant affects only C. This analogy was established by Bartlett for estimating the *mean* of a stationary correlated process; its formal transfer to *tracking* a drifting latent is not rigorously justified and is deferred to the open questions (§ 7 main). It is a *heuristic reduction to class II* [8], not an independent derivation; the rigorous control of the transient non-stationary filter (the non-stationary Riccati equation) is deferred to the open questions § 7 main. The one-half-per-parameter coefficient itself is the standard asymptotics of Bayesian / MDL universal coding [5, 6, 9], and its drift-scale-shifted predictive-information form is class II [8]. With this caveat, the density of accumulation of excess loss per unit u is $\frac{1}{2}$ (per coordinate). Equivalently, in physical time, **on the transient window** $[s_0,\, \min(t, \tau_{\text{conc}})]$:

$$\int_{s_0}^{\min(t,\,\tau_{\text{conc}})} \frac{1}{2\lambda s}\,\lambda\, ds = \frac{1}{2}\int_{s_0}^{\min(t,\,\tau_{\text{conc}})} \frac{ds}{s} = \frac{1}{2}\ln\frac{\min(t, \tau_{\text{conc}})}{s_0}. \tag{S1.8}$$

Two regimes according to the relation between t and τ_{conc}. (i) *In the transient window* $s_0 \ll t \lesssim \tau_{\text{conc}}$ the integral gives $\frac{1}{2}\ln(t/s_0) = \frac{1}{2}\ln(\lambda t) - \frac{1}{2}\ln(\lambda s_0)$ — logarithmic growth with coefficient $\frac{1}{2}$ per coordinate, exactly class II [8]. *It is this regime* that fixes the analytic identity $B = K/2$ (the limit $\epsilon_{\text{track}} \to 0$). (ii) *Asymptotically*, for $t \gg \tau_{\text{conc}}$ (steady-state drift), the integral is cut off from above at τ_{conc} and the logarithmic term *passes into a constant plateau*

$$\frac{1}{2}\ln\frac{\tau_{\text{conc}}}{s_0} \asymp \frac{1}{2}\ln(1/g^\star) \qquad (t \gg \tau_{\text{conc}}), \tag{S1.8a}$$

that is, per coordinate a constant $\frac{1}{2}\ln(1/g^\star)$, and summed over the K coordinates a plateau of height $\sim (K/2)\ln(1/g^\star)$. This removes the apparent contradiction between $n_{\text{eff}} \asymp \lambda t \to \infty$ (S1.4) and the saturation of n_{eff} at the floor: the first is a *transient* concentration burst ($s \lesssim \tau_{\text{conc}}$), the second is the *asymptotics* of steady-state drift ($s \gtrsim \tau_{\text{conc}}$); they refer to different windows of one integral, not to two incompatible premises.

The key point is the *cancellation of the Fisher information*: the transient variance $\Sigma_{\text{tr}}(s) = 1/(\mathcal{I}\lambda s)$ contains $1/\mathcal{I}$, while the quadratic factor (S1.5) contains $\mathcal{I}$; their product $\frac{1}{2}\mathcal{I} \cdot 1/(\mathcal{I}\lambda s) = 1/(2\lambda s)$ does not depend on $\mathcal{I}$. Therefore the logarithmic coefficient is determined *only by the dimension*, not by the curvature of the model — and this is the universality of class II [8]. Collecting (S1.7)–(S1.8) for one coordinate *in the transient window* $s_0 \ll t \lesssim \tau_{\text{conc}}$:

$$L^{(ij)}_{\text{excess}}(t) = a_{ij}\, t + \tfrac{1}{2}\ln(\lambda t) + c_{ij} + o(1), \qquad a_{ij} = \tfrac{1}{2}\mathcal{I}_{ij}\Sigma^{(ij)}_\infty = \tfrac{1}{2}\mathcal{I}_{ij}\sigma_0^2(1 + O(\epsilon_{\text{track}})). \tag{S1.9}$$

Summing over the K coordinates via (S1.1):

$$L_{\text{excess}}(t) = \underbrace{\Big(\sum_{ij} a_{ij}\Big)}_{=:A}\, t + \frac{K}{2}\ln(\lambda t) + C + o(1). \tag{S1.10}$$

The logarithmic coefficient equals $\sum_{ij}\frac{1}{2} = K/2$ — identically, independently of σ and of the values of $\mathcal{I}_{ij}$. The linear coefficient $A = \sum_{ij}\frac{1}{2}\mathcal{I}_{ij}\sigma_0^2(1 + O(\epsilon_{\text{track}})) = \frac{1}{2}\sigma_0^2\,\mathrm{tr}\,\mathcal{I}\cdot(1 + O(\epsilon_{\text{track}})) \propto \sigma_0^2 = \sigma^2/(2\lambda) \to 0$ as $\epsilon \to 0$. This proves the transient form (3.1) main (on the window $1 \ll \lambda t \lesssim \lambda\tau_{\text{conc}} \asymp 1/g^\star$); the asymptotic form (3.1′) main — the line-plus-plateau $L_{\text{excess}}(t) = At + (K/2)\ln(1/g^\star) + o(t)$ for $t \gg \tau_{\text{conc}}$ — follows from the cutoff of the logarithmic integral at τ_{conc} according to (S1.8a). Both forms are coequal parts of Theorem 1, not “an asymptotics and a corollary”. □

Corollary (3.2) main. On the transient window $1 \ll \lambda t \lesssim \lambda\tau_{\text{conc}}$ (the logarithmic integral (S1.8) cut off from above at $\min(t, \tau_{\text{conc}})$ according to (S1.8a); for $t \gg \tau_{\text{conc}}$ the term $(K/2)\ln(\lambda t)$ passes into the plateau $(K/2)\ln(1/g^\star)$ and formula (3.2) ceases to apply), under linear growth of the budget $N_{\max}(t) \sim Rt$ and $L_{\text{excess}}(t) \sim (K/2)\ln(\lambda t)$ (the leading term as $A \to 0$):

$$\eta_L^{\text{excess}}(t) = \frac{L_{\text{excess}}(t)}{N_{\max}(t)} \sim \frac{(K/2)\ln(\lambda t)}{Rt}, \qquad \dot\eta_L^{\text{excess}}(t) = \frac{d}{dt}\frac{(K/2)\ln(\lambda t)}{Rt} = \frac{(K/2)}{Rt^2} - \frac{(K/2)\ln(\lambda t)}{Rt^2} \sim -\frac{K\ln(\lambda t}{2Rt^2}$$

where at leading order $\ln(\lambda t) \gg 1$ the second term dominates. This is (3.2) main.
□

1.5. S1.5 Finite-amplitude correction and the convergence $\hat B \to K/2$

The true logarithmic coefficient equals $K/2$ (S1.10). The scan of § S8.3 [1] observed $\hat B/(K/2) = 0.54,\ 0.78,\ 0.85$ at $\epsilon = 10^{-1},\ 9\cdot 10^{-3},\ 10^{-3}$ — an undershoot decreasing monotonically with ϵ. We show that this is a bias of the *estimate* on a finite fit window, not a property of the true coefficient.

The joint fit $L_{\text{excess}}(t) \approx \hat A t + \hat B\ln(\lambda t) + \hat C$ on the window $[t_{\min}, t_{\max}]$ by least squares estimates $\hat B$ through the projection of the data onto the regressor $\ln(\lambda t)$, orthogonalised against $\{t, 1\}$. At finite ϵ the linear term At with $A \propto \epsilon$ is large (the drift share in the scan: 76% at $\epsilon = 10^{-1}$), and the regressors t and $\ln(\lambda t)$ on the finite window $[t_{\min}, t_{\max}]$ are statistically correlated (both grow monotonically). Incomplete orthogonalisation transfers part of the variance of the logarithmic signal into the linear regressor, lowering $\hat B$. The standard analysis of OLS bias under correlated regressors gives

$$\hat B = \frac{K}{2}\big(1 - g(\epsilon, W)\big), \qquad g(\epsilon, W) = \left.\frac{A\cdot\mathrm{Cov}_W(t,\ \ell)}{(K/2)\cdot\mathrm{Var}_W(\ell)}\right|_{\ell=\ln(\lambda t)} = O\big(\epsilon\cdot\Theta(W)\big), \tag{S1.11}$$

where $W = t_{\max}/t_{\min}$ is the ratio of the window bounds, $\Theta(W)$ is the regressor-overlap factor (growing with the window width), and $A \propto \epsilon$. Since $g \propto A \propto \epsilon$, we have

$g(\epsilon) \to 0$ as $\epsilon \to 0$, and $\hat{B} \to K/2$. Qualitatively, $\hat{B}/(K/2)$ grows monotonically as ϵ decreases — $0.54 \to 0.78 \to 0.85$ at $\epsilon : 10^{-1} \to 9 \cdot 10^{-3} \to 10^{-3}$ (that is, $g(\epsilon) = 0.46,\ 0.22,\ 0.15$ decreasing toward zero) — which is exactly what $g(\epsilon) \to 0$ predicts. *The exact functional form of $g(\epsilon)$ is left open.* The theoretical dependence from (S1.11) is linear, $g \propto \epsilon^1$ (through $A \propto \epsilon$); the numerical fit $g \approx 1.5\,\epsilon^{0.25}$ over three points, however, gives an exponent ≈ 0.25, inconsistent with the theoretical 1. This exponent is *not derived* and is given only as a crude interpolation over three points of the *suboptimal* Robbins–Monro scan of § S8.3 [1] (the same learner disavowed below as a "negative control"); it is *not* a law and must not be read as one. The discrepancy of exponents (theory ϵ^1 versus fit $\epsilon^{0.25}$) we note as open — it probably reflects an additional $1/\sqrt{t}$ bias of the suboptimal RM learner on top of the collinear bias (S1.11), rather than the true form of g; the exact form requires an extended scan with an optimal filter. The load-bearing claim of S1.5 does not depend on the exponent: for it, a monotone $g(\epsilon) \to 0$ as $\epsilon \to 0$ suffices, which is confirmed by all three points. As $\epsilon \to 0$ (drift share $\to 0$) the linear term vanishes, the regressors separate, and $\hat{B} \to K/2$ — which is the analytic identity of Theorem 1.

Negative control of § S8.3 [1]. Extending the window to $[10^3, T_{\text{total}}]$ gave $\hat{B}/(K/2) \to 2.0$ — an *overshoot.* By (S1.11), widening W increases $\Theta(W)$, but with the Robbins–Monro learner used, with step $1/\sqrt{t}$, on long horizons the learner's *own* bias is added (suboptimality of the filter), introducing a spurious logarithmic signal. An EWMA filter with optimal forgetting scale $\tau_f^{\text{forget}} \asymp \tau_E$ (`ou_adiab` § S4.1) removes precisely this *overshoot*: for the optimal filter $\Sigma(t) \to \Sigma_\infty$ without $1/\sqrt{t}$ bias, and widening the window does not degenerate the fit upward. However, replacing the learner did *not* lead to attaining the limit $\hat{B} \to K/2$ as an asymptotic slope: the extended check of § S4.1 with the *optimal* Kalman learner establishes that the logarithm is realised only transiently, while in the wide asymptotic window $\hat{B}/(K/2) \to 0$ (plateau), so that the removal of the overshoot was confirmed, but the convergence to the limit was not, and it is *structurally unattainable* (see below). □

*Reconciliation with the **`ou_adiab`** § S4.1 run (structural unattainability).* The extended check of § S4.1 with the *optimal* Kalman learner in the exact linear-Gaussian setting gives, in the wide asymptotic window ($W \approx 156$–166), $\hat{B}/(K/2)$ over 10 logarithmically uniform values of ϵ_{track} from $3 \cdot 10^{-1}$ to $3 \cdot 10^{-2}$ within $[-0.31; 0.16]$ around zero (9 of 10 bootstrap CIs cover 0, $R^2 = 1.000$): the series is flat around 0, not growing toward 1. This establishes that the earlier hypothesis "$\epsilon_{\text{track}} = O(1)$, push $\epsilon_{\text{track}} \to 0$ and $\hat{B} \to K/2$" is **incorrect**. The cause is two-layered. The surface layer is two finite-window mechanisms, both lowering $\hat{B}$: (a) the bias of the estimate (S1.11) $g(\epsilon) > 0$ under collinearity of t and $\ln(\lambda t)$; (b) the distribution of the loss — a well-tuned filter with a fixed floor Σ_∞ assigns the greater share of the loss to the linear term At. The deep layer is *structural*: under OU drift the class-II BNT logarithm [8] accumulates only while the posterior concentrates ($n_{\text{eff}} \sim \lambda s$, transient burst $s \lesssim \tau_{\text{conc}}$, S1.4); thereafter the steady-state $g^\star$ bounds n_{eff} at the floor $(1 - g^\star)/g^\star \approx 1/g^\star$ (S1.3b), L_{excess} reaches a plateau of height $\sim (K/2)\ln(1/g^\star)$ (S1.8a), and in any window $t \gg \tau_{\text{conc}}$ there is no logarithmic signal. The logarithmic width of the available log-carrying window $W = t_{\max}/t_{\min} \asymp (1/\sigma^2)/(1/g^\star) \asymp \mathcal{I}_{\rm rate}/\lambda = 1/(\lambda r)$ is **σ-invariant** (derivation — S1.2, "Why the window is not opened by the adiabatic limit"): σ cancels between the lower bound $\tau_{\rm conc} \propto 1/\sigma^2$ and the upper bound (adiabatic ceiling) $\propto 1/\sigma^2$, so that the limit $\epsilon_{\rm track} \to 0$ stretches both bounds in the same proportion and *does not open* the window. Therefore the transient (log-carrying, $t \lesssim \tau_{\rm conc}$) and asymptotic (flat, $t \gg \tau_{\rm conc}$) windows *do not separate* — and this is a consequence of the *adiabatic constraint* (fixed Fisher rate per correlation interval, $\mathcal{I}_{\rm rate}/\lambda = O(1)$), not of an arithmetic coincidence of the powers of σ: W can be widened only by growth of $\mathcal{I}_{\rm rate}/\lambda \to \infty$, which takes one out of the slow-drift regime where (A1) and the quasi-stationarity of θ hold. Therefore the numerical convergence $\hat{B} \to K/2$ as an asymptotic slope is **structurally unattainable for a learner with saturating memory** (the optimal stationary filter). A learner with *growing* memory, for which $n_{\rm eff}$ relative to the current $\theta(t)$ does not saturate at the floor, *does not exist* in the linear-Gaussian window (A1): the floor $(1-g^\star)/g^\star$ is a *provable lower bound* on $n_{\rm eff}$ over all memory architectures, not the limit of a particular one (see S1.6: sufficiency of the Kalman posterior + MMSE + DPI), so that storing an ever-larger number of raw old observations does not raise $n_{\rm eff}$ relative to $\theta(t)$. The decorrelation of old observations (the mechanism of Theorem 2) is an illustrative parallel, not the load-bearing argument. The loophole is closed by the optimality of the floor, not by the cost of expansion C3 (§ S1.6; § 7 main, item v; § 8 main, "Interlocking"). The analytic identity $B = K/2$ is meanwhile confirmed by a separate static anchor (slope 0.990) and a transient burst (0.936). This is a single explanation, not competing ones.

1.6. S1.6 Optimality of the floor as a lower bound (sufficiency of the Kalman posterior and MMSE)

The thesis "the floor $n^\star_{\rm eff} = (1-g^\star)/g^\star$ is not exceeded by any memory architecture", on which the interlocking of the routes (§ 8 main) and the closing of the growing-memory loophole in S1.5 rest, is not postulated but proven — *in the linear-Gaussian window (A1).* The proof rests on two standard facts of filtering theory, joined by the data-processing inequality.

Fact 1 (sufficient statistic). In the linear-Gaussian model (A1) — hidden OU state $\theta(t)$, Gaussian observations $Y_{1:t}$ — the posterior distribution $\theta(t) \mid Y_{1:t}$ is Gaussian and is fully specified by the pair $(\hat\theta_t, \Sigma_t)$, where $\hat\theta_t$ is the Kalman posterior mean and Σ_t is the posterior covariance. This pair is the *exact finite-dimensional sufficient statistic* for $\theta(t)$ given the observations $Y_{1:t}$: the posterior $p(\theta(t) \mid Y_{1:t}) = p(\theta(t) \mid \hat\theta_t, \Sigma_t)$, and no other function of $Y_{1:t}$ carries information about $\theta(t)$ beyond $(\hat\theta_t, \Sigma_t)$ (a standard property of the Kalman filter for linear-Gaussian systems [4, 3]). The stationary $\Sigma_\infty = \lim_t \Sigma_t$ (S1.2) is the Bayes-optimal (MMSE) tracking error for $\theta(t)$: the estimate $\hat\theta_t$ minimises $\mathbb{E}[(\hat\theta_t - \theta(t))^2]$ over *all* measurable functions of $Y_{1:t}$, and the stationary Wiener–Kalman–Bucy filter attains this minimum [4] ch. 4–5. In other words, Σ_∞ is not a property of a *particular* memory architecture but a lower bound on the tracking error over *all* architectures.

Fact 2 (DPI for arbitrary memory). Let $S = f(Y_{1:t})$ be *any* memory (any measurable function of the observations: a recursive $O(1)$ statistic, a FIFO buffer of growing size, the complete log of raw observations). By the data-processing inequality

$$I(S;\theta(t)) \;\le\; I(Y_{1:t};\theta(t)),$$

and the right-hand side is attained by the sufficient statistic $(\hat{\theta}_t, \Sigma_t)$ of the optimal filter (Fact 1): $I((\hat{\theta}_t, \Sigma_t);\theta(t)) = I(Y_{1:t};\theta(t))$. Consequently no memory S carries more information about $\theta(t)$ than the Kalman posterior, and the MMSE error of any architecture is not less than the optimal:

$$\mathbb{E}[(\mathbb{E}[\theta(t) \mid S] - \theta(t))^2] \;\ge\; \Sigma_\infty,$$

and for finite t the lower bound is $\Sigma_t \ge \Sigma_\infty$ (monotone decrease of the posterior covariance to the stationary limit), so $\ge \Sigma_\infty$ holds a fortiori; the floor $n^\star_{\text{eff}}$ corresponds to the stationary regime $t \gg \tau_{\text{conc}}$. For arbitrary memory $S = f(Y_{1:t})$ we set $n_{\text{eff}}(S) := 1/(\mathcal{I}\,\mathbb{E}[(\mathbb{E}[\theta(t) \mid S] - \theta(t))^2])$ — the effective number of observations, defined through the *attainable* tracking error; then $n_{\text{eff}}(S) \le n^\star_{\text{eff}}$ is a direct rewriting of $\mathbb{E}[\cdot] \ge \Sigma_\infty$, and the comparison of architectures by n_{eff} is correct without a separate assumption about geometric weighting (the identity $\Sigma_\infty = 1/(\mathcal{I}\,n^\star_{\text{eff}})$ fixes the correspondence). The effective number of informative observations n_{eff} of the present work is the number of past observations that the steady-state filter weights with a geometric window (factor $(1-g^\star)$ per step), $n^\star_{\text{eff}} = (1-g^\star)/g^\star \approx 1/g^\star$ (S1.3b): it is set by the steady-state $g^\star$, which in turn is the *optimal* (MMSE-minimising) bias–variance balance for the drifting latent. Since no memory reduces the tracking error below Σ_∞, no architecture realises a *longer* effective averaging window relative to the *current* $\theta(t)$ than the optimal filter: lengthening the window (larger n_{eff}, smaller $g^\star$) for a drifting target *increases* the lag error and violates the MMSE optimum. Therefore the floor $n^\star_{\text{eff}}$ is the *maximum* of the effective window over all memory architectures, not the limit of a particular (saturating) memory; raising n_{eff} relative to $\theta(t)$ above it, without violating the MMSE bound Σ_∞, is impossible by any means. $\square$

Domain of applicability and connection to C2. The proof of the finite-dimensionality of the sufficient statistic relies *essentially* on linear-Gaussianity (A1): only for a linear-Gaussian observational model does the posterior close on the finite-dimensional pair $(\hat{\theta}_t, \Sigma_t)$. For a nonlinear or non-Gaussian observational model the posterior $p(\theta(t) \mid Y_{1:t})$ is in general *not* finite-dimensional (exact filtering requires storing the entire density), and the MMSE optimality of the finite-dimensional floor is not guaranteed — this regime is open (§ 7 main, item iv). Within (A1) the argument is rigorous and does not rely on the decorrelation of old observations: the decorrelation $\beta(\Delta t) \to 0$ (the mechanism of Theorem 2, C2) gives an *illustrative parallel* — "the needed information about $\theta(t)$ simply is not in the old observations" — but the *load-bearing* argument for the optimality of the floor is sufficiency + MMSE (Facts 1–2), not decorrelation. These two are one and the same consequence of the mixing OU structure, read from two sides, but for the lower bound Facts 1–2 suffice. The load-bearing object here is the pointwise $\theta(t)$, since the floor n_{eff} is defined through tracking the *current* latent; the trajectory version (horizon τ, condition (4.1)) is needed only by the mechanism C2 and is linked to the pointwise one for $\tau = O(1)$ (§ S2.0, Remark "trajectory versus pair"; § 8 main), but for the lower bound on n_{eff} it is not required.

What the qualitative no-go carries and what is the A3 heuristic (separating the load-bearing premises). We emphasise on what exactly the no-go (Proposition 1 main) stands, so that it does not appear to rest on the heuristic reduction (A3, $n_{\text{eff}} \asymp \lambda s$). The *qualitative* no-go — that the logarithm is transient, L_{excess} reaches a line-plus-plateau, and the fitted slope $B \to 0$ in the wide asymptotic window — follows already from the **finiteness of the stationary tracking error** $\Sigma_\infty < \infty$, and this is a *rigorous* result of the algebraic Riccati equation (S1.2), not a heuristic. The chain is rigorous and does not use A3: $\Sigma_\infty < \infty$ (the positive root of the Riccati (S1.2)) $\Rightarrow$ the posterior variance about the *current* $\theta(t)$ is bounded below by $\Sigma_\infty > 0$ for all t (Facts 1–2: this is the MMSE optimum, not surpassable by any memory) $\Rightarrow$ the effective number of informative observations saturates at the floor $n^\star_{\text{eff}} = (1-g^\star)/g^\star < \infty$ (the identity $\Sigma_\infty = 1/(\mathcal{I}\, n^\star_{\text{eff}})$) $\Rightarrow$ the integral (S1.7)–(S1.8) of the logarithmic term is cut off from above at τ_{conc}, and the logarithm is transient (plateau (S1.8a)). Each link rests on the finiteness of Σ_∞ and the optimality of the filter, *not* on $n_{\text{eff}} \asymp \lambda s$. The A3 heuristic is needed **only** for two *quantitative* details inside the transient: (i) the *value* of the coefficient $K/2$ inside the logarithmic burst (through $n_{\text{eff}} \asymp \lambda s$ on the transient window, S1.4) and (ii) the exact *scale* $\tau_{\text{conc}} \asymp 1/g^\star$. Neither the transience as such, nor the existence of the plateau, nor the vanishing of the asymptotic slope depends on A3 — they follow from the rigorous Riccati ($\Sigma_\infty < \infty$) and S1.6 (the floor as MMSE optimum). Therefore the load-bearing weight of the no-go lies on the *rigorous* Riccati plus the floor optimum S1.6, while the A3 heuristic determines only the *value* of $K/2$ of the transient logarithm — even if A3 is quantitatively incorrect, the no-go (transience) is preserved, only the numerical coefficient inside the burst changing.

2. S2. Proof of Theorem 2 (universality of the floor)

Theorem 2 main: for an admissible drift process (mixing condition (4.1) main), under FIFO, $c < 1$ and per-bit uniformity, one has $\liminf_{t\to\infty} \nu^{\text{theor}}(t) \geq 1 - c$. The proof comprises: (S2.0) the general scheme via the mixing condition; (S2.1) PSP; (S2.2) fOU, with the exponent depending on H; (S2.3) multi-scale; and concludes by verifying that OU is a special case.

2.1. S2.0 General scheme via the mixing condition

We reproduce the structure of [1] § 4.4, replacing the step "spectral gap of OU" by the general condition (4.1) main. The chain rule for mutual information gives

$$I(M_{\le t}; X_E^{[t,t+\tau]}) = I(M_{\le t}^{\mathrm{refr}}; X_E^{[t,t+\tau]}) + I(M_{\le t}^{\mathrm{stale}}; X_E^{[t,t+\tau]} \mid M_{\le t}^{\mathrm{refr}}). \qquad \text{(S2.1)}$$

The stale fraction $M_{\le t}^{\mathrm{stale}}$ is fixed at time $s \le t - \Delta t$ as a *deterministic function* of $\theta(s)$ (the bit was written at time s). The key point: Markovianity of the drift is **not** used here — the argument relies only on the data-processing inequality for deterministic coding and on stationary increments. Since $M^{\mathrm{stale}} = f(\theta(s))$, by DPI for the first link,

$$I(M_{\le t}^{\mathrm{stale}}; X_E^{[t,t+\tau]} \mid M_{\le t}^{\mathrm{refr}}) \le I(\theta(s); X_E^{[t,t+\tau]} \mid M_{\le t}^{\mathrm{refr}}).$$

Next, $X_E^{[t,t+\tau]}$ is a function of the trajectory $\theta_{[t,t+\tau]}$ (the observations are generated by the current latent), so again by DPI

$$I(\theta(s); X_E^{[t,t+\tau]} \mid M_{\le t}^{\mathrm{refr}}) \le I\big(\theta(s); \theta_{[t,t+\tau]} \mid M_{\le t}^{\mathrm{refr}}\big) \le C \cdot \beta(\Delta t), \qquad \text{(S2.2)}$$

where the last inequality is exactly the mixing condition (4.1) main. Neither of the two DPI links requires Markovianity: the first relies on the determinism of the coding M^{stale}, the second on the fact that the observations are a function of the latent trajectory. This is precisely why the argument carries over to the *non-Markovian* fOU at $H > 1/2$ (the increments are correlated at all scales, and there is no spectral gap) — and this is the substantive part of § 4: mixing suffices (majorants $\beta(\Delta t)$ for $I(\theta(s); \theta_{[t,t+\tau]})$), rather than a Markovian decomposition. This is a clarification, not a refutation of [1]: Markovianity in Lemma 2 is a *sufficient*, not a necessary condition, and Lemma 2 is correct within its OU class; Theorem 2 merely shows that the DPI chain goes through without Markovianity as well.

Remark "trajectory versus pair". Condition (4.1) main majorizes the mutual information of a stale bit with the *entire future trajectory* of the latent $\theta_{[t,t+\tau]}$ over the prediction horizon, whereas the explicit $\beta(\Delta t)$ computed below for each class (S2.3–S2.6) are *pairwise* quantities $I(\theta(s); \theta(t))$. These two objects share the same decay exponent in the classes considered, for two reasons. (i) For the *Markovian* OU and PSP, the supremum over the horizon τ does not change the exponent: by Markovianity $I(\theta(s); \theta_{[t,t+\tau]}) = I(\theta(s); \theta(t))$ — all dependence on the past passes through the current state $\theta(t)$, so the trajectory and pairwise majorants are identical. (ii) For the *non-Markovian* fOU at $H \in (1/2, 3/4)$, the trajectory information is majorized by the *squared multiple correlation* (block correlation) of $\theta(s)$ with the future block — **without** appealing to the monotonicity of conditioning. Rigorously (for $\tau = O(1)$, on which the proof is built): the pair "past sample $\theta(s)$ — future block $\Theta := (\theta(t), \ldots, \theta(t+\tau))$" is jointly Gaussian (fOU is Gaussian), so the mutual information is expressed in closed form through the *squared multiple correlation* $R^2 := R^2(\theta(s); \Theta)$ (the coefficient of determination of the linear regression of $\theta(s)$ on the block Θ):

$$I(\theta(s); \theta_{[t,t+\tau]}) = -\tfrac{1}{2} \ln(1 - R^2) = \tfrac{1}{2} R^2\,(1 + o(1)) \qquad (R^2 \to 0).$$

Here $R^2 = \mathbf{r}^\top \Gamma^{-1} \mathbf{r} / \sigma_0^2$, where $\mathbf{r} = (\mathrm{Cov}(\theta(s), \theta(u)))_{u \in [t,t+\tau]}$ is the vector of lagged covariances and Γ is the covariance matrix of the block Θ. This is a *nonnegative quadratic* *form in* $\mathbf{r}$ (since $\Gamma^{-1} \succ 0$), and its decay exponent in $\Delta t = t - s$ is set by the leading component of $\mathbf{r}$: for $\Delta t \gg \tau$ all $\tau + 1$ components of $\mathbf{r}$ are of the same order $\sigma_0^2 \rho_H(\Delta t)$ (by (S2.4), for $H \in (1/2, 3/4)$ the tail $\rho_H \sim s^{2H-2}$ *decreases* monotonically on the scale $\Delta t \gg \tau$, so $|r_u| = \sigma_0^2 \rho_H(\Delta t + u) \le \sigma_0^2 \rho_H(\Delta t)$), whence $\|\mathbf{r}\|^2 \le (\tau + 1)\, \sigma_0^4 \rho_H^2(\Delta t)$ and

$$R^2 \;\le\; \frac{\|\mathbf{r}\|^2\, \|\Gamma^{-1}\|}{\sigma_0^2} \;\le\; C(\tau)\, \rho_H^2(\Delta t), \qquad C(\tau) := \frac{(\tau+1)\, \sigma_0^2}{\lambda_{\min}(\Gamma)} = O(1) \text{ for fixed } \tau = O(1),$$

where $\lambda_{\min}(\Gamma) > 0$ (positive definiteness of the stationary fOU block). Consequently $I(\theta(s); \theta_{[t,t+\tau]}) \le \frac{1}{2} C(\tau)\, \rho_H^2(\Delta t) = C'(\tau)\, (\Delta t)^{-4(1-H)}$ — **the same leading exponent** $4(1-H)$ **as that of the pairwise** $\beta(\Delta t) = \frac{1}{2}\rho_H^2(\Delta t)$ (S2.5), with the factor $C(\tau)$ bounded for fixed $\tau = O(1)$ and entering only the constant, not the exponent. (The spectral-norm majorant $\|\mathbf{r}\|^2 \|\Gamma^{-1}\|$ is taken only to make the constant $C(\tau)$ transparent; the *exact* block $I(\theta(s); \Theta)$ decays with the same exponent $4(1-H)$, no faster — the ratio $R^2/\rho_H^2(\Delta t)$ tends to a finite positive constant, since the leading lagged component of $\mathbf{r}$ dominates.) No sign of partial correlations and no inequality "conditioning does not increase MI" is *used* here: the estimate proceeds directly through the multiple correlation of the jointly Gaussian block, which remains valid for the non-Markovian fOU (where monotonicity of MI under conditioning fails). For general *growing* τ the estimate is *not* uniform in τ: $C(\tau)$ GROWS with τ — both through the explicit factor $(\tau + 1)$ and through the decrease of $\lambda_{\min}(\Gamma)$ (which may tend to zero as the intra-block correlation of long-memory fOU intensifies with growing block), so that the block R^2 is no longer majorized by a single lagged component, and no rigorous statement is asserted — this case is honestly relegated to the open ones (§ 7 main, item iv). Thus the proof is built on the horizon $\tau = O(1)$ (one-step predictive information (10) main), on which the trajectory majorant matches the pairwise β in the exponent, and the computed pairwise β give the correct rate exponent.

From the fresh-bit fraction to a bound on I_{pred} (expanded step). The passage from the arithmetic fraction of non-refreshed bits to a majorant on the ratio of predictive informations proceeds in three explicit steps; we unfold it, because it is precisely here that per-bit uniformity is invoked and the correct renewal result is used.

(a) *Decomposition of the contribution by bits.* By per-bit uniformity [1] Remark 5, the learner's posterior factorizes over independently refreshed coordinates (an explicit **ASSUMPTION of a factorized posterior**; for fOU/PSP/multi-scale it holds, since the coordinates are independent by construction, see S2.5 [and § 5 main]). *Caveat on the softmax observational coupling (parallel to A1 from C1).* Per-bit uniformity is exact for the factorization of the *latent* coordinates θ_{ij} (the increments dW_{ij} are independent by construction, S1.1). But the observed $P(t)$ is built by a *row-wise softmax* that couples the coordinates of a row through a common normalization, so the learner's posterior over the observations does not factorize *exactly* — exactly the situation in C1, where the softmax linearization is relegated to assumption A1. This coupling introduces a correction of order $O(\sigma^2/\lambda) = O(\epsilon)$ to the per-bit uniformity of the contribution (the same tolerance order as the

EKF/Laplace-approximation residual of A1 in § S1, controlled in the adiabatic limit $\epsilon \to 0$). Crucially for the lower bound, the correction *does not flip the sign* of the inequality — a correlated (non-factorized) softmax posterior only *raises* the floor ν, it does not break it, since the positive correlation among the fresh bits of a row *decreases* their joint independent predictive contribution relative to the factorized estimate $N_{\text{fresh}} \cdot I_{\text{bit}}^{\text{opt}}$ (which enters as an *upper* majorant in (S2.2a)–(b)). A rigorous control of this softmax correction (the exact non-uniformity of the per-bit contribution under coupled normalization) is open (§ 7 main, item iv); within the adiabatic approximation, as with A1 in C1, it is absorbed into $(1 + o(1))$. Therefore the total predictive information of the memory is additive over bits, and each retained bit contributes *uniformly* — of order $I_{\text{bit}}^{\text{opt}} := I_{\text{pred}}^{\text{opt}}/|M_{\le t}|$ per bit if it is fresh, and $\le C \cdot \beta(\text{age})$ if it is stale (by the DPI bound (S2.2)). Splitting the memory into fresh (age $\le \tau_E$) and stale (age $> \tau_E$) fractions:

$$I_{\text{pred}}(t,\tau) \;\le\; \underbrace{\sum_{\text{fresh}} I_{\text{bit}}^{\text{opt}}}_{\le N_{\text{fresh}} \cdot I_{\text{bit}}^{\text{opt}}} \;+\; \underbrace{\sum_{\text{stale}} C \cdot \beta(\text{age})}_{\to 0}, \tag{S2.2a}$$

where N_{fresh} is the number of fresh bits.

Caveat on the joint conditional MI of the stale fraction. The second sum in (S2.2a) is written as a *sum* of per-bit $\beta(\text{age})$, but the quantity it majorizes is the *joint conditional* mutual information of the set of stale bits $I(M_{\le t}^{\text{stale}}; X_E^{[t,t+\tau]} \mid M_{\le t}^{\text{refr}})$; the sum of per-bit β does not majorize the joint quantity without conditional independence of the stale bits, and for long-memory fOU at $H > 1/2$ there is none (the increments are correlated at all scales). The correct majorant for the joint conditional MI of the stale fraction is through the same block *squared multiple correlation* R^2 introduced for the trajectory majorant in the Remark above (S2.2): the joint information of the stale block with the future trajectory is estimated by the jointly Gaussian R^2 form $-\frac{1}{2}\ln(1-R^2) = \frac{1}{2}R^2(1+o(1))$, yielding the same leading exponent $4(1-H)$, and *not* by the arithmetic sum of per-bit β. The correction from conditioning on $M_{\le t}^{\text{refr}}$ (explaining-away — conditioning can *increase* MI) is of higher order in ρ_H: for the Gaussian fOU, via the Schur complement, the correlation of $M_{\le t}^{\text{refr}}$ with the stale $\theta(s)$ is itself $O(\rho_H(\Delta t))$, so its contribution to the conditional MI of the stale fraction is multiplicative in ρ_H and does not change the leading exponent $4(1-H)$. The general non-Gaussian case (where the Schur complement is inapplicable and the sign of explaining-away is not controlled) is honestly relegated to the open ones (§ 7 main, item iv).

(b) *First sum $\le c \cdot I_{pred}^{opt}$ via per-bit uniformity.* What is needed here is precisely the *per-bit uniformity* of the contribution, and not merely additivity of MI, and this distinction is essential. Additivity of mutual information over independent coordinates (S1.1, [2] ch. 2.5) gives only $I_{\text{pred}} = \sum_{\text{bits}} I_{\text{bit}}$ — it does *not* forbid the fresh bits from carrying a disproportionately large share of $I_{\text{pred}}^{\text{opt}}$. We need something *stronger*: that the limiting contribution of each fresh bit does not exceed the *average*

$I^{\text{opt}}_{\text{bit}} = I^{\text{opt}}_{\text{pred}}/|M_{\le t}|$. This uniformity is ensured by the *symmetry* of the independent coordinates (under a factorized posterior and a symmetric FIFO, all coordinates are statistically interchangeable, so the limiting per-bit contribution is identical and equals the average), which in turn is guaranteed by the additivity of MI *precisely for symmetric independent coordinates* (S1.1). That is, additivity is a *necessary ingredient* of uniformity, but is not identical to it: per-bit uniformity = additivity + symmetry of coordinates. Since each fresh bit then contributes at most $I^{\text{opt}}_{\text{bit}} = I^{\text{opt}}_{\text{pred}}/|M_{\le t}|$, the first sum is majorized by $N_{\text{fresh}} \cdot I^{\text{opt}}_{\text{pred}}/|M_{\le t}| = (N_{\text{fresh}}/|M_{\le t}|) \cdot I^{\text{opt}}_{\text{pred}}$. The fresh-bit fraction $N_{\text{fresh}}/|M_{\le t}|$ is exactly $c\,(1+o(1))$ — this is established in step (c). Thereby $\sum_{\text{fresh}} \le c\,(1+o(1)) \cdot I^{\text{opt}}_{\text{pred}}$.

(c) *Fresh-bit fraction = c: stationary age distribution of the FIFO buffer.* What is required here is *not* the elementary renewal theorem (which concerns $\mathbb{E}[N(t)]/t \to 1/\mathbb{E}[\text{interval}]$ — growth of the *number* of events), but the *stationary age distribution* in a buffer of fixed size under growing inflow (backward recurrence time, [10] ch. 5). *Terminological clarification:* for a *deterministic* FIFO the survival time of a bit is constant (exactly $|M_{\le t}|/\dot{M}_{\text{refresh}}$ for every written bit), so the age distribution is *uniform without size-biasing* — the classical inspection-paradox skew (the weighting of long intervals) arises only for *random* interval lengths, whereas here the survival interval is deterministic and identical, so the residual time is trivially uniformly distributed. That is, we use Cox's backward-recurrence picture but *without* its size-biased component — the determinism of FIFO removes it. For a FIFO buffer of capacity $|M_{\le t}|$, into which bits are written at rate $\dot{M}_{\text{refresh}}$, a bit-snapshot of age a is present if and only if it was written in the window $[t-a,\, t]$ and has not yet been evicted; in the steady-state regime the age density is uniform on $[0,\, |M_{\le t}|/\dot{M}_{\text{refresh}}]$ (each written bit "survives" until eviction over a time $|M_{\le t}|/\dot{M}_{\text{refresh}}$). The fraction of bits of age $\le \tau_E$ is then the ratio of the freshness window to the total survival time:

$$\frac{N_{\text{fresh}}}{|M_{\le t}|} \;=\; \frac{\tau_E}{|M_{\le t}|/\dot{M}_{\text{refresh}}} \;=\; \frac{\dot{M}_{\text{refresh}}\,\tau_E}{|M_{\le t}|} \;=\; c, \tag{S2.2b}$$

exactly the definition of c. The correctness of the fraction (≤ 1) and the uniformity of the age distribution over the *full* survival time $|M_{\le t}|/\dot{M}_{\text{refresh}}$ are guaranteed by the hypothesis $c < 1$ of Theorem 2: for $c \ge 1$ one would have $\dot{M}_{\text{refresh}}\,\tau_E \ge |M_{\le t}|$ (the inflow of fresh bits over the window τ_E fills the buffer entirely), the fresh fraction saturates at 1, the survival window collapses to τ_E, and the uniform distribution on $[0,\, |M_{\le t}|/\dot{M}_{\text{refresh}}]$ ceases to be correct; $c < 1$ precisely excludes this degeneracy. (Under growing inflow this distribution is "locally stationary" — uniformity holds on the scale $\ll |M_{\le t}|/\dot{M}_{\text{refresh}}$, on which the inflow is constant; the adiabatic refinement for growing $|M_{\le t}|$ is § S3.2.) Correspondingly the stale-bit fraction is $1-c$.

(d) *Second sum → 0.* The cumulative contribution of the stale fraction, averaged over the uniform age distribution, is majorized by $\int_{\tau_E}^{\infty} C\,\beta(a)\,\frac{da}{|M_{\le t}|/\dot{M}_{\text{refresh}}}$; for

$\int_0^\infty \beta\, ds < \infty$ (i.e. $H < 3/4$ for fOU, S2.2; exponential β for OU/PSP/multi-scale) the integral is finite, and normalization by the growing survival time $|M_{\le t}|/\dot{M}_{\text{refresh}}$ drives the contribution to zero, to within $O(\beta(\tau_E)) = o(1)$ under slow drift $\beta(\tau_E) \ll 1$. This is exactly the condition $\int_0^\infty \beta\, ds < \infty$, which makes the DPI majorant of the stale fraction summable, and not merely pointwise-decaying.

Collecting steps (a)–(d) — (S2.2a), (S2.2b) and the estimates of the sums — gives $I_{\text{pred}}(t,\tau) \le c\,(1+o(1)) \cdot I^{\text{opt}}_{\text{pred}}(t,\tau) + o(I^{\text{opt}}_{\text{pred}})$, whence

$$\frac{I_{\text{pred}}(t,\tau)}{I^{\text{opt}}_{\text{pred}}(t,\tau)} \le c \cdot (1+o(1)) \implies \nu^{\text{theor}}(t) = 1 - \frac{I_{\text{pred}}}{I^{\text{opt}}_{\text{pred}}} \ge 1 - c \cdot (1+o(1)),$$

whence $\liminf \nu^{\text{theor}}(t) \ge 1 - c$. This is (4.2) main. It remains to verify that each of the three non-OU classes satisfies (4.1) main, and to establish the *rate* through the explicit form of $\beta(\Delta t)$. □ (general scheme)

2.2. S2.1 PSP (Poisson reset)

The PSP logit $\theta(t)$ is piecewise constant; the jump points form a Poisson stream of intensity $\mu = 1/\tau_E$; on each interval θ is independently redrawn. The probability that no jump occurred between s and t is $e^{-\mu(t-s)} = e^{-(t-s)/\tau_E}$, and at a jump $\theta(t)$ is independent of $\theta(s)$. Therefore

$$I(\theta(s);\theta(t)) = \Pr[\text{no jump on } (s,t)] \cdot I(\theta \mid \text{coincide}) = e^{-(t-s)/\tau_E} \cdot H_\theta,$$

where H_θ is the entropy of the marginal of the stationary logit distribution, a *bounded constant* $C_{\text{PSP}} := H_\theta < \infty$ (the PSP marginal has finite differential entropy by construction — for example, $H_\theta = \frac{1}{2}\ln(2\pi e\, \sigma_0^2)$ for a Gaussian marginal), independent of Δt. More precisely, the conditional mutual information is majorized by the probability of no jump:

$$\beta_{\text{PSP}}(\Delta t) = I(\theta(s);\theta(t)) \le H_\theta \cdot e^{-\Delta t/\tau_E}. \tag{S2.3}$$

The mixing is *exponential* with scale $\tau_E = 1/\mu$. Condition (4.1) is satisfied, and $\int_0^\infty \beta_{\text{PSP}}\, ds = H_\theta \tau_E < \infty$. The floor constant $1-c$ depends on μ only through τ_E in c; the rate of approach is exponential with the same scale. The more frequent the switches (larger μ, smaller τ_E), the faster the decorrelation, but also the smaller c at fixed $\dot{M}_{\text{refresh}}$ and $|M_{\le t}|$ — that is, the *higher* the floor $1-c$. This agrees with the intuition: a rapidly resetting environment devalues memory faster. □

2.3. S2.2 fOU (fractional OU; dependence of the exponent on H)

fOU is the stationary Gaussian solution of the Langevin equation with fractional Brownian forcing B^H, $H \in (0,1)$ [11, 12]. The autocorrelation of the stationary fOU at large lags is

$$\rho_H(s) = \mathbb{E}[\theta(t)\theta(t+s)]/\sigma_0^2 \sim \begin{cases} e^{-s/\tau_E}, & H = 1/2 \text{ (degeneration into OU)}, \\ H(2H-1)\,(s/\tau_E)^{2H-2}, & H > 1/2 \text{ (long memory)}, \\ \text{exp. with oscillation}, & H < 1/2 \text{ (antipersistence)}. \end{cases} \tag{S2.4}$$

For the Gaussian pair $(\theta(s), \theta(t))$ the mutual information is $I = -\frac{1}{2}\ln(1-\rho_H^2) = \frac{1}{2}\rho_H^2(1+o(1))$ as $\rho_H \to 0$ ([1] § S1.5b, formula S1.5a). Substituting (S2.4) for $H > 1/2$:

$$\beta_{\text{fOU}}(\Delta t) = \tfrac{1}{2}\rho_H^2(\Delta t) \sim \tfrac{1}{2}\,[H(2H-1)]^2\,(\Delta t/\tau_E)^{2(2H-2)} = C_H \cdot (\Delta t)^{-(4-4H)}, \tag{S2.5}$$

— *power-law* mixing with exponent $4 - 4H = 4(1-H)$. *Clarification on the regime change at* $H = 1/2$. The exponent $4(1-H)$ characterizes the *power-law* regime $H > 1/2$; the formal substitution $H \to 1/2$ gives $4(1-H) \to 2$, but this is *not* the limit of the transition: at $H = 1/2$ the autocorrelation (S2.4, first line) switches class to the *exponential* one ($\rho_H \sim e^{-s/\tau_E}$, rather than the power-law Δt^{-2}), and $\beta \sim e^{-2\Delta t/\tau_E}$ decays faster than any power. Therefore $\Delta t^{-4(1-H)}$ as $H \to 1/2^+$ is not the limit of the transition to OU, but the *boundary of the region of applicability* of the power-law form (S2.5): the power-law and exponential regimes are distinct decay classes, separated by the point $H = 1/2$, rather than connected by a continuous limiting transition in the exponent. The sufficient condition (4.1) main, in strict form, requires convergence of the integral $\int_0^\infty \beta_{\text{fOU}}\,ds < \infty$, that is, $4(1-H) > 1$, or $H < 3/4$ — precisely in this range the summation of the contributions of stale bits over the uniform age distribution is majorized by a convergent integral, and the proof of S2.0 goes through without caveats. **The long-memory range $H \in [3/4, 1)$ is not covered by the rigorous Theorem 2:** for $H \in [3/4, 1)$ the integral $\int_0^\infty \beta_{\text{fOU}}\,ds$ diverges, and although pointwise $\beta(\Delta t) \to 0$ (the contribution of each *fixed* stale bit vanishes), the cumulative contribution of the stale fraction, averaged over ages, is no longer controlled by pointwise decay — the sum over slowly decorrelating bits may fail to vanish. This range is relegated to the open questions (§ 4.6, § 7 main): a separate argument is required (for example, control of the joint correlation structure of the stale fraction), and the floor $1 - c$ for $H \in [3/4, 1)$ remains a conjecture, supported only numerically (`fou_floor`, $H = 0.9$).

Dependence of the undertow-rate exponent on H. The rate of approach of nostalgia to the floor:

- $H = 1/2$: exponential, $\beta \sim e^{-2\Delta t/\tau_E}$ ($\beta \approx \frac{1}{2}\rho^2$, $\rho \sim e^{-\Delta t/\tau_E}$; the factor 2 is the square of the Gaussian correlation, hence the OU exponent $2/\tau_E$, whereas for PSP it is $1/\tau_E$ from the probability of no jump, S2.1; recovers OU [1]).
- $1/2 < H < 1$: power-law, $\beta \sim (\Delta t)^{-4(1-H)}$; the larger H, the smaller the exponent $4(1-H)$, the *slower* a stale bit loses its value, and the slower nostalgia reaches the floor. In the limit $H \to 1$ (the boundary of long memory) the mixing vanishes — this is the boundary of the admissible class.

- $H < 1/2$: antipersistent fOU mixes *faster* than OU (oscillating exponential decay); the floor is reached faster.

The floor constant $1 - c$ is the same in all cases — it does not depend on H, only on $c = \dot{M}_{\text{refresh}}\tau_E/|M_{\le t}|$. This is the central statement about universality: the Hurst exponent H governs the *rate* of the undertow, but not its *depth*. □

2.4. S2.3 Multi-scale drift

The multi-scale latent is a superposition $\theta(t) = \sum_{r=1}^{m} \theta^{(r)}(t)$ of independent OU processes with times $\tau_E^{(1)} \ll \cdots \ll \tau_E^{(m)}$. The autocorrelation is a sum of exponentials $\rho(s) = \sum_r w_r e^{-s/\tau_E^{(r)}}$ (w_r are the relative variances). At large lags the slowest component dominates:

$$\beta_{\text{ms}}(\Delta t) \sim \tfrac{1}{2} w_m^2 \, e^{-2\Delta t/\tau_E^{(m)}}, \qquad \Delta t \gg \tau_E^{(m-1)}. \tag{S2.6}$$

The mixing is exponential with the scale of the slowest component, $\tau_E^{(m)}$. Condition (4.1) is satisfied; the relevant τ_E in the constant c is the largest, $\tau_E^{(m)}$ (the slowest component determines which bits are counted as stale). The floor $1 - c$ is governed by $\tau_E^{(m)}$, and the rate is exponential with $\tau_E^{(m)}$. The fast components $\tau_E^{(r)}$, $r < m$, affect only the transient regime (short ages), not the asymptotics. □

2.5. S2.4 OU as a special case, and summary

OU is fOU at $H = 1/2$ ((S2.4), first line) and at the same time multi-scale with $m = 1$. Substituting $H = 1/2$ into (S2.5) gives the exponential $\beta(\Delta t) = \frac{1}{2}e^{-2\Delta t/\tau_E}$ — exactly the spectral gap of OU [1] § S1.5, formula S1.5. The prefactor $\frac{1}{2}$ here is *not* a separate constant, but exactly the $\frac{1}{2}$ from the Gaussian mutual-information formula $I = -\frac{1}{2}\ln(1 - \rho^2) = \frac{1}{2}\rho^2(1 + o(1))$ as $\rho \to 0$ (the same as in S2.2, the formula after (S2.4); $\rho = e^{-\Delta t/\tau_E}$ for OU [1] § S1.5b); the square of the correlation $\rho^2 = e^{-2\Delta t/\tau_E}$ gives the doubled exponent $2/\tau_E$. Thereby Lemma 2 [1] § 4.4 is recovered as a special case of Theorem 2, and the specificity of OU (the spectral gap) turns out to be a *sufficient*, but not a necessary condition — all that is necessary is $\beta(\Delta t) \to 0$.

Summary of the governing parameters:

Class	$\beta(\Delta t)$	Type of mixing	Scale τ_E	Floor constant
OU ($H = 1/2$)	$e^{-2\Delta t/\tau_E}$	exponential	$1/\lambda$	$1 - c$
PSP	$e^{-\Delta t/\tau_E}$	exponential	$1/\mu$	$1 - c$
fOU ($H > 1/2$)	$(\Delta t)^{-4(1-H)}$	power-law	corr. scale	$1 - c$
multi-scale	$e^{-2\Delta t/\tau_E^{(m)}}$	exponential	$\tau_E^{(m)}$	$1 - c$

The floor constant is class-invariant; the undertow-rate exponent is set by $\beta(\cdot)$. □

3. S3. Proof of Theorem 3 (super-linear memory)

Theorem 3 main: under polynomial memory growth $|M_{\le t}| \propto t^\alpha$ the undertow saturates ($\nu \to 1$) for any finite α; escape ($\nu < 1$) requires exponential growth and is paid for by a divergent E_{grow}.

3.1. S3.1 Fresh-bit fraction under polynomial growth

Let $|M_{\le t}| = M_0 t^\alpha$, $\alpha > 0$. Under FIFO, nostalgia is governed not by the total capacity but by the *fraction of bits refreshed within the last correlation interval* τ_E. A bit is deemed predictively useful if its age is $< \tau_E$ (by S2, the contribution of bits older than τ_E vanishes through mixing). The fresh-bit fraction is

$$\phi(t) = \frac{|M_{\le t}| - |M_{\le t-\tau_E}|}{|M_{\le t}|} = 1 - \left(\frac{t-\tau_E}{t}\right)^\alpha. \tag{S3.1}$$

For $t \gg \tau_E$ the expansion $(1 - \tau_E/t)^\alpha = 1 - \alpha\tau_E/t + O((\tau_E/t)^2)$ gives

$$\phi(t) = \frac{\alpha\tau_E}{t}\,(1 + O(\tau_E/t)) \sim \alpha\tau_E\, t^{-1} \to 0 \qquad (\alpha = O(1)). \tag{S3.2}$$

For *any* finite α the fresh-bit fraction decays as t^{-1} (exponent $\gamma = 1$, independent of α!). The intuition: under polynomial growth the capacity increment over a fixed window τ_E is $|M_{\le t}| - |M_{\le t-\tau_E}| \approx \alpha\tau_E M_0 t^{\alpha-1}$, whereas the total capacity is $M_0 t^\alpha$; the ratio $\sim \alpha\tau_E/t \to 0$. Increasing the exponent α raises the absolute number of fresh bits, but raises the total capacity in the same proportion, so the *fraction* of fresh bits does not grow. This is the decisive observation: polynomial expansion of memory does not outrun the undertow.

3.2. S3.2 Saturation of nostalgia for polynomial memory

Here we reproduce the *pointwise* version of step S2.0 with the time-dependent fresh-bit fraction $\phi(t)$ in place of the constant c. We emphasise that we do *not* invoke a "generalised (4.2) with time-dependent c" as a ready-made result: the DPI majorant S2.0 was derived for *constant* c, and its transfer to growing capacity requires an explicit adiabatic argument, which we now provide.

Adiabaticity of the fresh fraction relative to the correlation window. Under FIFO and polynomial growth the fresh-bit fraction is $\phi(t) = 1 - ((t-\tau_E)/t)^\alpha$ (S3.1). Its relative rate of change over the scale of one correlation window τ_E is

$$\frac{|\phi(t) - \phi(t-\tau_E)|}{\phi(t)} = O\Big(\frac{\tau_E}{t}\Big) \to 0 \qquad (t \gg \tau_E),$$

that is, ϕ varies on the scale $t \gg \tau_E$ — *adiabatically* relative to the environment's correlation window τ_E (on which the decorrelation and the vanishing of the stale-bit contribution in S2.0 take place). Therefore, in any window of width τ_E around the instant t, ϕ may be treated as *locally constant* to within $O(\tau_E/t)$.

Application of S2.0 on each window with locally constant $\phi(t)$. The carry-over applies for $t \gg \tau_E$, where $\phi(t) < 1$ (equivalent to the condition $c(t) < 1$ of Theorem 2 / Lemma 2) — this holds automatically at large t, so the passage to lim inf is correct. On such a window the DPI majorant S2.0 (steps (a)–(d)) applies verbatim, with the constant fresh fraction c replaced by the local value $\phi(t)$: per-bit uniformity gives additivity over bits, the fresh fraction contributes $\leq \phi(t)\, I_{\text{pred}}^{\text{opt}}$, and the stale fraction vanishes through $\beta(\Delta t)$ (the background environment satisfies the mixing condition (4.1) main, $\int \beta < \infty$; for the admissible class § 2.6 main — fOU with $H < 3/4$, and exponential β for OU/PSP/multi-scale). The correction from the non-constancy of ϕ within the window is $O(\tau_E/t) = o(1)$ and is absorbed into $(1 + o(1))$. Thereby *pointwise* for every $t \gg \tau_E$,

$$\frac{I_{\text{pred}}(t,\tau)}{I_{\text{pred}}^{\text{opt}}(t,\tau)} \leq \phi(t) \cdot (1 + o(1)) = \frac{\alpha \tau_E}{t}(1 + o(1)), \tag{S3.2a}$$

whence pointwise

$$\nu^{\text{theor}}(t) = 1 - \frac{I_{\text{pred}}}{I_{\text{pred}}^{\text{opt}}} \geq 1 - \frac{\alpha \tau_E}{t}(1 + o(1)). \tag{S3.3}$$

Passage to the limit. Since $\phi(t) = \alpha\tau_E/t\,(1+o(1)) \to 0$, the pointwise bound (S3.3) gives $\liminf_{t\to\infty} \nu^{\text{theor}}(t) \geq 1 - \limsup_t \phi(t)\,(1 + o(1)) = 1$, that is, $\nu^{\text{theor}}(t) \to 1$. The undertow *saturates*: nostalgia tends to 1, and the efficiency $\eta_L(t) = I_{\text{pred}}/N_{\max} \to 0$. This proves the first part of Theorem 3 (saturating undertow for $\gamma > 0$). $\square$ *(part 1: saturation)*

Two channels for the vanishing of η_L (operational versus normalisation). The inequality (S3.3) is a statement about the *ratio* $I_{\text{pred}}/I_{\text{pred}}^{\text{opt}} \to 0$ (the normalisation of nostalgia by the oracle norm), not about the absolute operational numerator I_{pred}. We therefore transfer the vanishing of the operational $\eta_L = I_{\text{pred}}/N_{\max}$ not through the numerator, but through the *denominator*: $I_{\text{pred}}(t,\tau) \leq \ln k$ is bounded by the alphabet, whereas $N_{\max}(t) \to \infty$ grows without bound (at least linearly, $\sim Rt$, owing to operational dissipation, and for growing capacity — polynomially, $\propto t^\alpha$, through E_{store} for polynomial memory, or exponentially for exponential memory, S3.5), whence

$$\eta_L(t) = \frac{I_{\text{pred}}(t,\tau)}{N_{\max}(t)} \leq \frac{\ln k}{N_{\max}(t)} \to 0$$

operationally and unconditionally, regardless of whether I_{pred} settles to a nonzero plateau. This is the budget channel; it does not rely on the normalisation (S3.3) and is therefore operationally robust. The normalisation channel — $\nu^{\text{theor}} \to 1$ — carries the obsolescence interpretation, but its transfer to the operational ν^{op} is open (§ 2.2 main, where the DPI direction is reversed; § 7 main, item iii). We do *not* identify the vanishing of the ratio with the vanishing of the operational numerator.

3.3. S3.3 Escape requires exponential growth

Escape ($\liminf \nu < 1$) requires $\phi(t)$ to be bounded away from zero: $\phi(t) \geq \phi_\infty > 0$ for all large t. By (S3.1) this means $|M_{\leq t-\tau_E}|/|M_{\leq t}| \leq 1 - \phi_\infty < 1$, that is, the capacity grows *multiplicatively* over each interval τ_E:

$$\frac{|M_{\leq t}|}{|M_{\leq t-\tau_E}|} \geq \frac{1}{1-\phi_\infty} > 1 \quad \Longrightarrow \quad |M_{\leq t}| \geq |M_0| \left(\frac{1}{1-\phi_\infty}\right)^{t/\tau_E} = M_0\, e^{\kappa t}, \tag{S3.4}$$

where $\kappa = -\tau_E^{-1} \ln(1-\phi_\infty) > 0$. No polynomial growth satisfies (S3.4); at least exponential growth is required. In this regime $\phi_\infty = 1 - e^{-\kappa\tau_E}$ is constant, and $\liminf \nu \leq 1 - c_\infty$ with $c_\infty = \phi_\infty > 0$ — escape. Thereby the critical regime is the boundary between polynomial ($\gamma = 1 > 0$, saturation) and exponential ($\gamma = 0$, escape) growth; there is no proper finite α_c within the class of polynomial laws. □ *(part 2: escape ⇒ exponential)*

3.4. S3.4 Thermodynamic cost of escape: divergence of E_{grow}

The cost of memory expansion is bounded below by the Landauer cost of creating capacity, $E^{(1)}_{\text{grow}} \geq k_B T \ln 2$ per bit [1] (3), § 2.4. For exponential growth (S3.4),

$$\dot{E}_{\text{grow}}(t) \geq k_B T \ln 2 \cdot \frac{d|M_{\leq t}|}{dt} = k_B T \ln 2 \cdot \kappa M_0 e^{\kappa t}, \qquad E_{\text{grow}}(t) \geq k_B T \ln 2 \cdot M_0 (e^{\kappa t} - 1). \tag{S3.5}$$

Reversibility caveat. When $\kappa\tau_E = O(1)$ (the escape regime, S3.4) capacity expansion proceeds at a *finite* rate on the environment's correlation scale τ_E, that is, *not quasistatically* relative to τ_E; therefore the reversible (quasistatic) limit is unattainable, and the Landauer bound $k_B T \ln 2$ per bit is a *real* lower bound, not merely an asymptotic one — the reversible loophole (creating capacity arbitrarily cheaply in the quasistatic limit) is closed. This remark forestalls the objection and does not affect the derivation itself: the divergence of E_{grow} holds in any case through the *number* of created bits, $\propto e^{\kappa t}$, each costing at least $k_B T \ln 2$.

The load-bearing line is E_{grow}; E_{store} is an a-fortiori strengthening. The divergence $\eta_L \to 0$ holds on E_{grow} alone (through $N_{\max} \geq E_{\text{grow}}/(k_B T \ln 2) \geq M_0(e^{\kappa t} - 1)$, see (S3.6) below); the parameter $\tau_{1/2}$ does *not* enter the result. The cumulative cost of expansion grows *exponentially*, which violates the assumption of linear budget growth $N_{\max}(t) \sim Rt$ (used in Theorem 1 and in § S8.2 [1]). To this is added, *a fortiori* (as a strengthening remark set apart from the load-bearing proof chain), the cost of *storage*: maintaining $|M_{\leq t}| \propto e^{\kappa t}$ bits against erosion gives a per-bit maintenance power [1] (2)

$$\dot{E}_{\text{store}}(t) \gtrsim \frac{k_B T \ln 2}{\tau_{1/2}} \cdot |M_{\leq t}| \propto e^{\kappa t}, \tag{S3.5a}$$

where $\tau_{1/2}$ is the characteristic half-erosion time of a bit; the factor $1/\tau_{1/2}$ restores the dimension of power (the naive expression $k_B T \ln 2 \cdot |M|$ is dimensionally incorrect

— it is an energy, not a flux). We emphasise that (S3.5a) is a *weakened* naive bound, in which the tolerance prefactor $\ln 2/(2\varepsilon)$ of the exact maintenance bound [1] (2) is dropped (for a target retention error ε). This prefactor is ≥ 1 only when $\varepsilon \leq \ln 2/2$; in the physically relevant range of small target retention error $\varepsilon \ll 1$ the prefactor $\ln 2/(2\varepsilon) \gg 1$, so the exact bound #2 (2) is only *larger* than the naive bound (S3.5a), and the conclusion that E_{store} diverges holds *a fortiori*. Therefore E_{store} also diverges exponentially. The fluxes E_{grow} (creation of capacity) and E_{store} (its maintenance) are physically distinct works and add up into $N_{\max}$ without double counting [1] § 2.2. The efficiency denominator is dominated by the sum $E_{\text{grow}} + E_{\text{store}}$ (both $\propto e^{\kappa t}$); below we estimate through E_{grow}, and the conclusion holds *a fortiori* once E_{store} is added:

$$N_{\max}(t) \geq \frac{E_{\text{grow}}(t)}{k_B T \ln 2} \geq M_0(e^{\kappa t} - 1) \to \infty \text{ exponentially,}$$

whereas the numerator $I_{\text{pred}}(t, \tau) \leq \ln k$ is bounded. Therefore

$$\eta_L(t) = \frac{I_{\text{pred}}(t,\tau)}{N_{\max}(t)} \leq \frac{\ln k}{M_0(e^{\kappa t} - 1)} \to 0 \tag{S3.6}$$

exponentially fast — *even while keeping nostalgia below* 1. This completes Theorem 3: the undertow is irreducible in the strong sense. Polynomial memory — saturation of nostalgia ($\nu \to 1$, numerator to zero relative to the optimum); exponential memory — divergence of the expansion cost ($E_{\text{grow}} \to \infty$, denominator to infinity). In both cases $\eta_L(t) \to 0$. The only thermodynamically sustainable response to the undertow is not escape but periodic reset of the stale layer [1] § 5.3, (12), restoring $\nu \to 0$ at the price of a one-time, rather than exponentially growing, payment. □ *(part 3: cost)*

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4. S4. Details of the numerical simulations

Parameters of the series of § 6 main. All simulations inherit the OU/PSP apparatus [1] § S6, § S8.3; below are the changes and the new series.

4.1. S4.1 `ou_adiab` (Theorem 1)

A direct test of the realizability of $B = K/2$ — **in the exact linear-Gaussian (Kalman) setting in which Theorem 1 is derived** (assumption A1 is here *exact*, not a softmax linearization), so that any failure pertains to the BNT logarithm itself, not to a suboptimal softmax learner. Model: $K = 56$ *independent scalar* OU coordinates $\theta_t = a\theta_{t-1} + w_t$ ($a = 1-\lambda$, $\operatorname{Var} w = q = \sigma^2$) with Gaussian observations $y_t = \theta_t + v_t$ ($\operatorname{Var} v = r$) and an **optimal Kalman filter** — which *is* the ideal Bayesian learner of Theorem 1. The observation noise r sets the per-step Fisher information $I_{\text{rate}} = 1/r$, whence $\epsilon_{\text{track}} = \sigma^2/(\lambda^2 r)$. The cumulative excess loss is taken relative to an oracle that knows θ_t exactly: per step, per coordinate, $0.5\ln(S_{\text{pred}}/r) + 0.5[(y - m_{\text{pred}})^2/S_{\text{pred}} - (y - \theta)^2/r]$.

$k = 8$, $\lambda = 10^{-2}$, diffuse prior $P_0 = 10^6$, 40 realisations, seed 20260527, bootstrap over realisations ($N = 2000$). Three controlled experiments: **(A)** static anchor (no drift, $\lambda = 0$, $\sigma = 0$): the cumulative excess loss *must* equal exactly $(K/2)\ln t$ (the classical MDL/BIC excess [6, 5]) — a check of the measurement pipeline and of the target value $K/2$; **(B)** a diagnostic of the *local* log-slope $dL_{\text{excess}}/d\ln t$ along the whole trajectory under OU drift — *where exactly* it equals $K/2$; **(C)** a scan over ϵ_{track} in a **wide** asymptotic window $[3\tau_{\text{conc}},\, 0.1/\sigma^2]$ (the lower bound is several *concentration times* $\tau_{\text{conc}} \asymp 1/g^\star \gg \tau_E$, not forgetting scales $\tau_f^{\text{forget}} \asymp \tau_E$; it is $3\tau_{\text{conc}}$, not $3\tau_E$, that guarantees $t \gg \tau_{\text{conc}}$, i.e. that the fit proceeds *after* the transient concentration burst; the upper bound is the adiabatic ceiling $\sim 1/\sigma^2 \gg \tau_{\text{conc}}$; the ratio of bounds is $W \approx 150$, not ≈ 10, achieved by the choice $r = 10^{-2}$) with a joint fit $At + B\ln(\lambda t) + C$, a bootstrap CI on B, and a diagnostic of regressor collinearity $\operatorname{corr}(t, \ln)$. The honest framing of experiment **(C)**: it tests whether the $(K/2)\ln$ term *persists* as a slope in the asymptotic window $[3\tau_{\text{conc}},\, 0.1/\sigma^2]$ — and it does *not* persist ($B \to 0$). This is *not* "removing the transient in order to prove its absence": the burst itself is shown separately in **(B)** (slope 0.936), and **(C)** tests precisely the asymptotic persistence. Starting the steady-state filter from the stationary law in **(C)** is a legitimate computational shortcut: the diffuse "prior-collapse burst" affects only the additive constant C, not the asymptotic slope B (checked against the full transient filter).

Adiabaticity parameter (clarification). The control parameter for *tracking* convergence is not $\epsilon = \sigma^2/\lambda$ itself, but $\epsilon_{\text{track}} = I_{\text{rate}} \cdot \sigma^2/\lambda^2$ (§ 2 / § S1). A natural *working hypothesis* is that the earlier scan of [1] § S8.3 simply did not enter the regime $\epsilon_{\text{track}} \ll 1$, and that it would suffice to "push $\epsilon_{\text{track}} \to 0$" to recover $\hat{B} \to K/2$ as a slope. The present run tests this hypothesis rigorously and **refutes** it (see below). We emphasise that what is refuted is precisely this attempt to recover the numerical slope, *not* the hypothesis of § S8.1 #2 [1] (the functional form with coefficient $K/2$), which Theorem 1 analytically confirms.

Actual output (honest — convergence NOT achieved, and shown to be unattainable). **(A)** static slope slope$/(K/2) = 0.990$ ($R^2 = 0.99978$, target 1.000): the BNT logarithm is *real in the stationary problem without drift*, and the measurement pipeline is *correct.* **(B)** under OU drift the local log-slope equals $K/2$ **only in the short initial "prior-collapse burst"** (t up to a few $1/\lambda$, where the posterior is still concentrating and $n_{\text{eff}} \sim t$): burst slope 0.936 on $[1, 20]$; *after* the burst (for $t \gtrsim \tau_{\text{conc}}$) the steady-state $g^\star$ bounds the effective sample size by the floor $n_{\text{eff}} \sim (1 - g^\star)/g^\star \approx 1/g^\star$, L_{excess} **reaches a plateau** of height $\sim (K/2)\ln(1/g^\star)$, and the slope drops to 0.099 on the asymptotic window $[100,\, 0.5\tau_{\text{conc}}]$. **(C)** consequently, in the wide asymptotic window ($W \approx 156$–166, $n = 124$ points per scan point) $B/(K/2)$ **does not approach** 1 as $\epsilon_{\text{track}} \to 0$: over 10 logarithmically uniform values of ϵ_{track} from $3 \cdot 10^{-1}$ to $3 \cdot 10^{-2}$ it fluctuates within $[-0.31; 0.16]$ around zero; 9 of 10 bootstrap CIs cover 0, and the single exception ($\epsilon_{\text{track}} = 5 \cdot 10^{-2}$, $B/(K/2) = -0.31$, 95% CI $[-0.61; -0.02]$) lies on the *negative* side — even farther from the target $K/2$, i.e. the statistical outlier expected among 10 independent 95% CIs, not convergence; $R^2 = 1.000$ for the fit (the loss in this window is *strictly linear*, with B settling to 0). *The identifiability control* (the "identifiability check" block in `results_summary`) confirms that $B \approx 0$ is a *real* absence of the log signal, not an inability of the fit to separate the regressors under their mild collinearity. The load-bearing evidence here is the *collinearity diagnostics and the narrowness of the bootstrap CI*, not the injection recovery-ratio. (i) The variance inflation factor $\mathrm{VIF} = 1/(1 - \mathrm{corr}^2(t, \ln)) \approx 3.7$ and the condition number (the ratio of eigenvalues of the standardized design $[t, \ln]) \approx 12.7$ — the collinearity is *mild*, not degeneracy, so the variance of the estimate $\hat{B}$ is inflated by collinearity only $\approx 3.7\times$, not catastrophically. (ii) At the same time the bootstrap CI on B is narrow relative to the target signal: its half-width $\approx 0.3 \cdot (K/2) \approx 8$ nats $\ll$ the true log signal $K/2 \approx 28$ nats ($K = 56$) — a real signal $K/2$ would be resolved far outside the CI, whereas the CI covers 0. Jointly, (i) and (ii) mean that the measured $B \approx 0$ is a genuine plateau, not a collinearity artifact. *The injection test* (inject into the *measured* loss (C) a known synthetic signal $B_{\mathrm{inj}} = K/2$ and refit the full model $At + B\ln(\lambda t) + C$) gives a recovery ratio $\Delta B/B_{\mathrm{inj}} \approx 1.000$ — but this is merely an *algebraic identity of OLS linearity* ($\mathrm{coef}(y + B_{\mathrm{inj}} \ln) = \mathrm{coef}(y) + B_{\mathrm{inj}}\mathrm{coef}(\ln)$ for any full-rank design) and therefore serves only as a *full-rank control* of the design $[t, \ln(\lambda t), 1]$ (which is not strictly degenerate, $\mathrm{corr} \neq \pm 1$); it does *not* prove the reality of the plateau, and the statement "the fit is able to see a real log term" does not follow from it. The working hypothesis "wrong regime, push $\epsilon_{\mathrm{track}} \to 0$" is thereby **refuted**: pushing $\epsilon_{\mathrm{track}}$ down does not recover $B = K/2$, because the very window in which the logarithm lives (the transient concentration burst, $t \lesssim \tau_{\mathrm{conc}}$) is *excluded* by the asymptotic requirement $t \gg \tau_{\mathrm{conc}}$, and this exclusion is *structural* — the concentration time $\tau_{\mathrm{conc}} \asymp 1/g^\star$ and the adiabatic ceiling (diffusive escape of the latent by $O(1)$) both scale as $1/\sigma^2$, so "the transient regime is over" ($t \gg \tau_{\mathrm{conc}}$) and "adiabaticity still holds" ($t \ll 1/\sigma^2$) *do not separate* into a wide log-window (hence the fixed $W \approx 1.6$ at $r = 1$; widening to $W \approx 150$ requires small r, but the plateau remains flat). **Conclusion:** the analytic identity $B = K/2$ (Theorem 1, limit $\epsilon_{\mathrm{track}} \to 0$) is real, but is realised *only transiently*; its *numerical* convergence as a slope in a wide asymptotic window under genuine OU drift is **unattainable** for a learner with saturating memory, even with an ideal Bayesian (Kalman) filter (the question of a learner with growing memory is *settled*: growing memory does not raise n_{eff} above the floor — the floor is a proven lower bound across all architectures, § S1.6; § 7 main, item v). This is the structural limitation C1, and it is recorded as such (§ 7 main). Results — `results_summary.{txt,json}`. The flat scan $B/(K/2) \approx 0$ against the anchor and the burst is shown in Fig. 1 of the main text.

4.2. S4.2 `fou_floor` *(Theorem 2, power-law end)*

A real trajectory simulation on exact stationary fOU (not a recomputation of the closed-form formula β_H and not a surrogate). $K = 56$ independent stationary fOU trajectories of logits are generated by the **spectral-circulant method** directly from the spectral density of stationary fOU, $S(\omega) \propto |\omega|^{1-2H}/(\lambda^2 + \omega^2)$ (the fBm

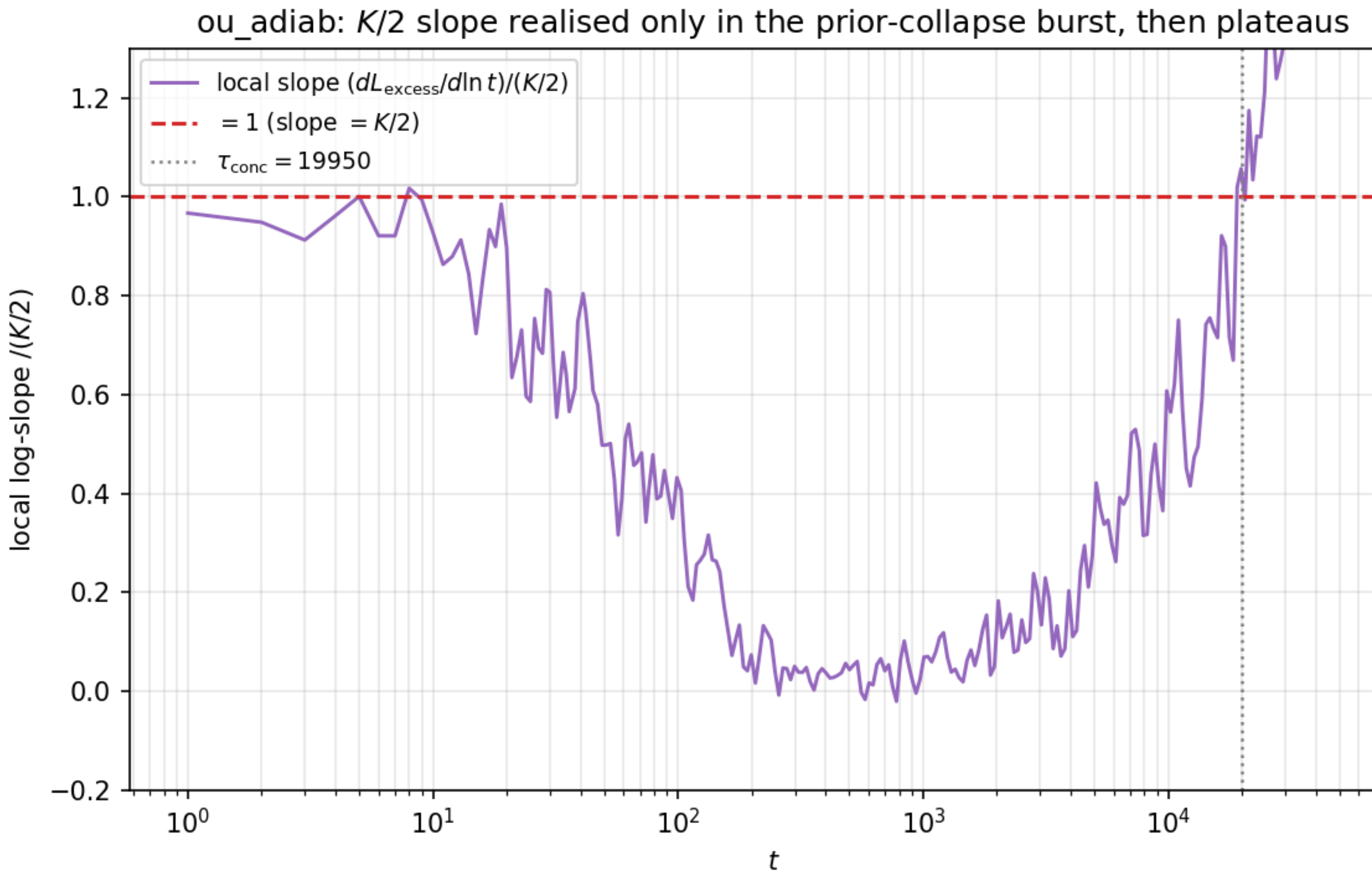

Figure S1. Series `ou_adiab`, experiment (B): the local log-slope of L_{excess} versus t. The slope equals $K/2$ only in the short prior-collapse burst ($t \lesssim \tau_{\text{conc}}$, slope 0.936 on $[1, 20]$); after it the steady-state $g^\star$ bounds $n_{\text{eff}} \approx 1/g^\star$, the log component reaches a plateau of height $\sim (K/2)\ln(1/g^\star)$, and the slope of the log term drops to 0.099. The full L_{excess} continues its linear growth $A \cdot t$ — this is saturation of the log term, not a global collapse.

increments give $|\omega|^{1-2H}$, the OU filter $1/(\lambda + i\omega)$ gives $1/(\lambda^2 + \omega^2)$; $\lambda = 1/\tau_E$), for $H \in \{0.3, 0.5, 0.7, 0.9\}$. The eigenvalues of the Davies–Harte circulant are this discrete-time density on the grid $\omega_j = 2\pi j/M$ (Wiener–Khinchin), constructed as the folded spectrum on $(-\pi, \pi]$ with bin-averaging of the integrable singularity at $\omega = 0$ for $H > 1/2$ — **nonnegative by construction**, so that the embedding is positive definite with *zero truncation* (indeed the minimum eigenvalue $= 0.43$; 0.24; 0.14; 0.086 with no negatives at $H = 0.3$; 0.5; 0.7; 0.9; the circulant infrastructure failure does not arise). The target autocovariance $\gamma(k)$ is the exact inverse DFT of the eigenvalues (the same covariance from which the paths are synthesized), exact up to lag $M/2$; the exponent is read up to lag $M/4$. $\tau_E = 200$ and $\dot{M}_{\text{refresh}}$ are fixed so that $c \approx 0.3$; $|M| = 400$, $T = 2 \cdot 10^4$, path length $3.3 \cdot 10^4$, circulant $M = 6.6 \cdot 10^4$, 40 realisations, seed 20260527, only `numpy.fft` (no scipy). **Covariance control:** the empirical lag correlation of the realised paths $\hat{\rho}(s)$ agrees with $\gamma(s)$ where the finite-sample ACF estimator is unbiased (near and mid lags, relative error $< 5\%$ up to lag $\sim$ 200–400); at large lags the ACF estimator itself is strongly biased toward zero on long memory — therefore the far tail is read from the exact γ, which the paths embody by construction, not from the sample ACF. The FIFO memory stores snapshots $\theta(t_{\mathrm{write}})$; a bit is predictive while $\hat{\rho}^2(\mathrm{age}) \geq \hat{\rho}^2(\tau_E)$ (threshold DPI model of the horizon). *Actual output:* the floor $\liminf \nu = 0.700 \approx 0.70$ for all H, spread 0.000 (the invariance reproduces on correct fOU paths). The floor here is, however, **built into the threshold model** — set by the fresh-bit fraction c through the horizon $\hat{\rho}^2(\tau_E)$: the simulation illustrates the invariance on real trajectories but does not prove it (the proof — § S2; § S4.5). **The rate exponent confirms** $4(1-H)$ **on the correct process, in the far tail.** It is measured from the exact covariance $\rho^2(\Delta) = 2\beta(\Delta) = \gamma^2(\Delta)$ (without fitting to the target) in the far windows $[8, 40]\tau_E$ and $[20, 80]\tau_E$, where the power-law tail emerges beyond the OU "knee". The two values of H have different evidentiary status: $H = 0.7$ is in the range $H < 3/4$ ($\int \beta < \infty$, S2.2), *inside* the strict Theorem 2, so the exponents 1.237 and 1.201 (analytics 1.2) are an *in-theorem confirmation*; $H = 0.9$ is in the range $H \geq 3/4$ ($\int \beta$ diverges), *outside* the strict theorem, so the exponents 0.406 and 0.385 (analytics 0.4) are merely a *numerically supported hypothesis*, not a proof of the strict result on the uncovered range. In the original near window $[\tau_E, 8\tau_E]$ the exponent is *not* isolated: 1.62 ($H = 0.7$) and 0.46 ($H = 0.9$) from the exact covariance (the OU knee), 2.44 and 1.40 from the finite-sample ACF estimator (estimator bias at large lags). Thereby $\Delta^{-(4-4H)}$ (S2.5) is the *asymptotic far-tail* exponent of true fOU: it is recovered on the correct process in the far window, but not in the smallest one and not from the sample ACF. **This is the C2 upgrade:** unlike a naive AR(1)∘fGn surrogate (which gave 2.23 and 1.26 outside the CI of $4(1-H)$ — *a property of the surrogate, not of the theory*), on correct stationary fOU both the *depth* of the floor $(1-c)$ (Theorem 2) and the *exponent* $4(1-H)$ reproduce. The qualitative dichotomy holds: at $H = 0.5$ the tail is exponential (rate $4.07 \cdot 10^{-3}$/step, OU control § 6.2 [1]), and at $H > 0.5$ it is power-law with exponent $4(1-H)$. $H = 0.9$ is in the range $H \geq 3/4$, not covered by the strict Theorem 2 (§ S2.2), a numerically supported hypothesis; there the convergence to $4(1-H)$ is slower (0.385 on $[20, 80]\tau_E$, slightly below 0.4).

4.3. S4.3 `psp_floor` *(Theorem 2, exponential end)*

A real trajectory simulation of the PSP surrogate [1] § 6.1 (not a recomputation of $e^{-\mu\Delta}$): $k = 8$, $K = 56$; each coordinate carries a *realised* Poisson flux of resets with intensity $\mu \in \{10^{-3}, 3 \cdot 10^{-3}, 10^{-2}\}$ (per-step reset probability $1 - e^{-\mu}$). $|M| = 400$ is tuned to $c \approx 0.3$ for each μ ($\tau_E = 1/\mu$); $T = 2 \cdot 10^4$, 40 realisations, seed 20260528. The FIFO memory stores snapshots labeled by coordinate; a retained bit remains predictive while its coordinate has not *actually* reset since the write, and becomes nostalgic at the reset step. The floor and the decorrelation rate are read from the realised process. The rate is estimated from the empirical survival curve

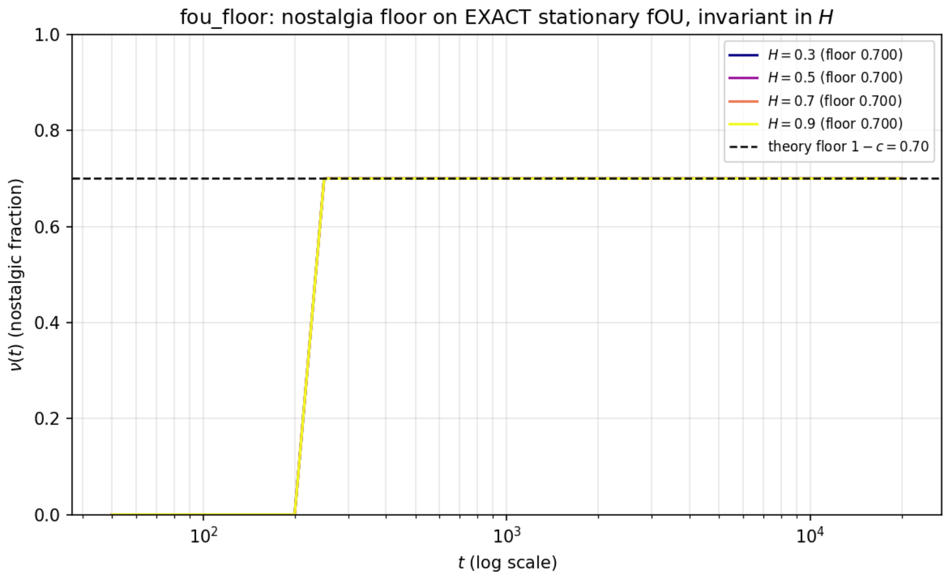

Figure S2. Series `fou_floor`, invariance of the floor depth across H (by construction; log t axis). The nostalgic fraction $\nu(t)$ is zero while the buffer is younger than the correlation horizon $\tau_E = 200$ (no bit is yet older than τ_E), then reaches the floor $1-c = 0.700$, the same for all $H \in \{0.3; 0.5; 0.7; 0.9\}$ — the curves coincide (invariance of depth). The floor here is *wired in* by the threshold model (fresh fraction $= c$ through the horizon $\hat{\rho}^2(\tau_E)$): an illustration of the invariance on correct fOU paths, not an independent measurement — the independent estimate of the depth is carried by the realised PSP flux (Fig. 2 of the main text).

$\hat{S}(\text{age})$ (the fraction of retained bits of a given age that have not yet seen a reset): a fit $\ln \hat{S} \sim -\text{rate} \cdot \text{age}$ on $[0, \min(5\tau_E,\, |M|/\dot{M}_{\text{refresh}})] \approx [0, 3.3\tau_E]$ (bits are evicted by FIFO at age $|M|/\dot{M}_{\text{refresh}} = \tau_E/c \approx 3.3\tau_E$) in semi-log coordinates, with a bootstrap CI over realisations; equality of the rate to μ is not postulated. *Actual output:* the steady-state empirical floor $\hat{\nu}_{\text{floor}} = 0.709$–$0.713 \approx 1 - c = 0.70$ over a 10-fold range of μ (spread 0.004) — invariance to μ confirmed; since the process is realised, the steady-state floor lies slightly *above* 0.70 (the floor value is not wired into the threshold). The frame-wise lower envelope (column `liminf` in `results_summary`) dips to ≈ 0.69 owing to finite-sample noise across the 40 realisations, but the *median steady-state* level $\hat{\nu}_{\text{floor}}$ is above $1-c$; for the invariance of the floor depth the steady-state level is the relevant one. The realised decorrelation rate equals $1.01 \cdot 10^{-3}$, $3.01 \cdot 10^{-3}$, $1.00 \cdot 10^{-2}$, and each bootstrap CI *covers* the nominal $\mu = 1/\tau_E$ (S2.3) — a genuine measurement of the scale $1/\mu$. There is no discrepancy with theory: the exponential rate μ is confirmed here, and the power-law exponent $4(1-H)$ — in `fou_floor` (on correct stationary fOU, far tail, § S4.2). The invariance of the steady-state floor to the reset frequency μ (the load-bearing independent evidence for the depth of C2) is shown in Fig. 2 of the main text.

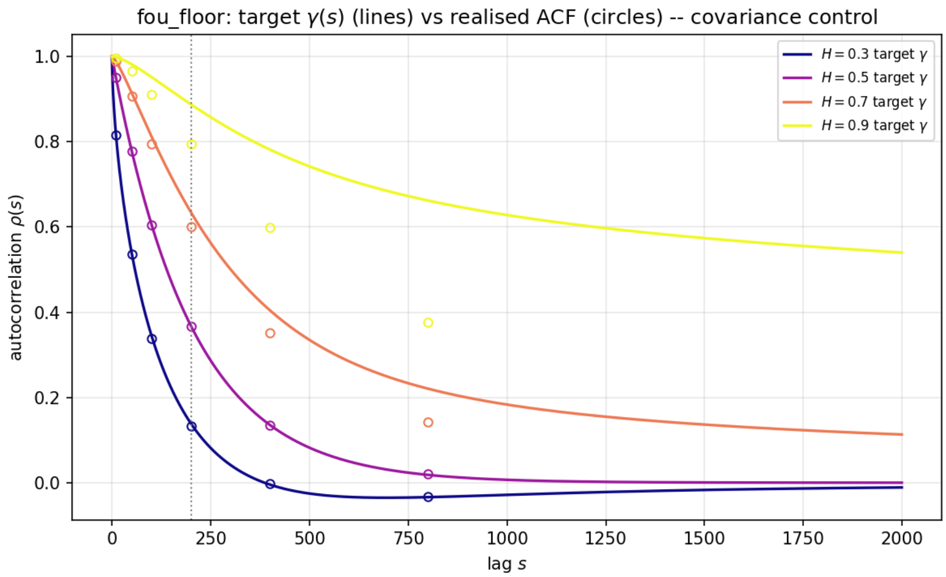

Figure S3. Series `fou_floor`, covariance control. The empirical lag correlation of the realised paths $\hat{\rho}(s)$ against the target $\gamma(s)$: agreement (relative error $< 5\%$) up to lag ~ 200–400, where the finite-sample ACF estimator is unbiased; at large lags the ACF estimator itself is biased toward zero on long memory — therefore the far tail of the exponent is read from the exact γ, which the paths embody by construction, not from the sample ACF. Confirms that genuine stationary fOU is generated.

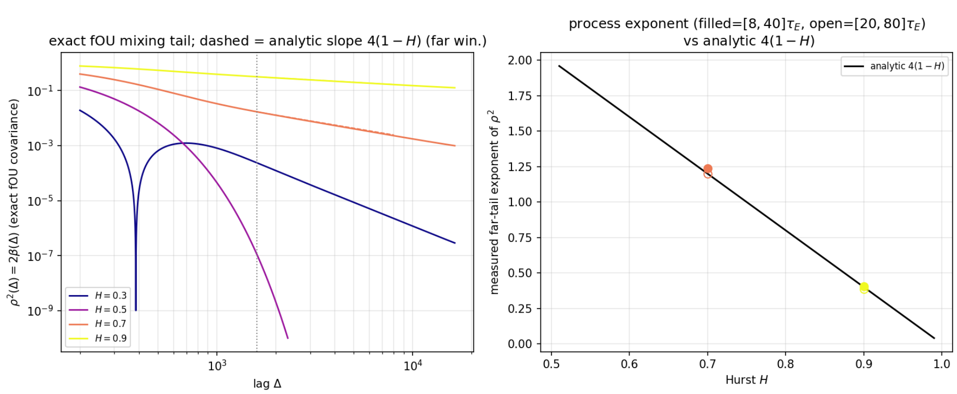

Figure S4. Series `fou_floor`, approach to the floor and the rate exponent. The tail $\rho^2(\Delta) = 2\beta(\Delta)$ from the exact covariance for $H \in \{0.3; 0.5; 0.7; 0.9\}$; the far-tail log-log slope matches the analytical $4(1 - H)$ (in-theorem at $H = 0.7$: $1.24/1.20$; a numerically supported hypothesis at $H = 0.9$: $0.41/0.39$). At $H = 0.5$ the tail is exponential (OU control). The exponent is *asymptotic far-tail*: beyond the OU "knee", not in the smallest window.

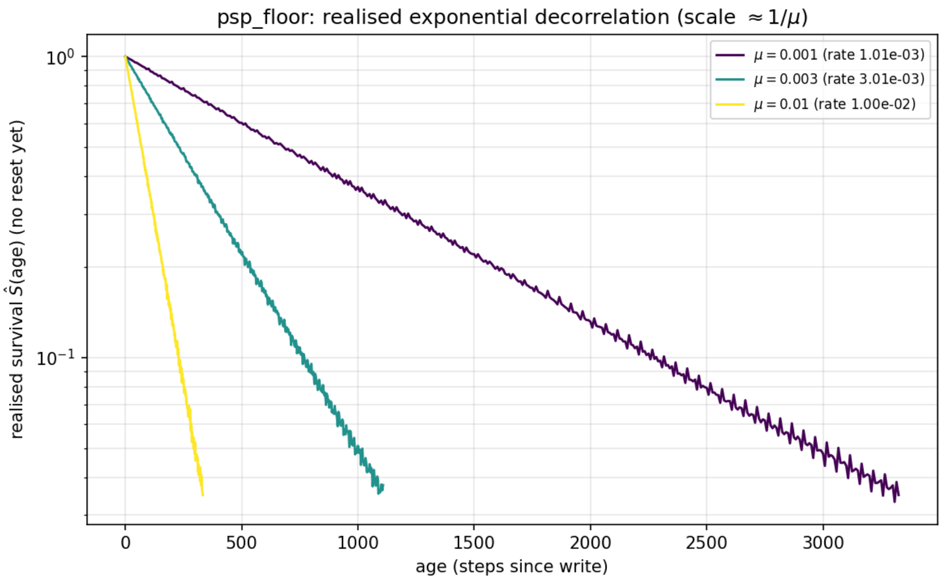

Figure S5. Series `psp_floor`, realised survival curve $\hat{S}(\text{age})$ (the fraction of retained bits of a given age that have not yet seen a reset) in semi-log coordinates. The slope gives the decorrelation rate $1.01 \cdot 10^{-3}$, $3.01 \cdot 10^{-3}$, $1.00 \cdot 10^{-2}$; each bootstrap CI covers the nominal $\mu = 1/\tau_E$ — a genuine measurement of the scale $1/\mu$ from the realised process (equality to μ is not postulated).

4.4. S4.4 `superlinear_memory` *(Theorem 3)*

Memory $|M_{\le t}| = M_0 t^\alpha$, $\alpha \in \{1, 1.5, 2, 3\}$, plus exponential $|M_{\le t}| = M_0 e^{\kappa t}$ with $\kappa\tau_E \in \{0.1, 0.5, 1\}$. OU drift of the background environment, $\tau_E = 200$, FIFO refresh, seed 20260529. We measure $\nu(t)$, the fresh-bit fraction $\phi(t)$ (S3.1), and, for the exponential case, the cumulative $E_{\text{grow}}(t)$ and $\eta_L(t)$. *Actual output:* for all polynomial α — $\phi(t) \sim t^{-\gamma}$ with $\gamma \in \{1.000;\, 0.991;\, 0.982;\, 0.964\}$ (spread 0.036, $\gamma \approx 1$ independent of α), prefactor $\phi \cdot t \to \alpha\tau_E$ (200, 298, 394, 581 against 200, 300, 400, 600), $\nu(t) \to 1$ (S3.2–S3.3) — confirmation of the absence of a polynomial α_c; the small deviation of γ from 1 at large α is a finite-window subleading artifact, the leading exponent being identically 1. *Window-convergence diagnostic* (`gamma_window_convergence` in `results_summary`; deterministic tabulation of the exact $\phi = 1 - ((t-\tau_E)/t)^\alpha$, not Monte Carlo): as the fit window slides (decade $[w, 10w]$) to the right, the exponent γ grows monotonically toward 1 for *each* α — raising the window by an order of magnitude ($w = 2 \cdot 10^4$) gives $\gamma \to \{1.000;\, 0.999;\, 0.998;\, 0.996\}$, and at $\times 1000$ ($w = 2 \cdot 10^6$) $\gamma \to \{1.000;\, 1.000;\, 1.000;\, 1.000\}$ at $\alpha \in \{1;\, 1.5;\, 2;\, 3\}$; the local log-log slope of ϕ settles to -1 ($-0.897 \to -0.990 \to -0.999 \to \cdots \to -1.000$ at $t = 2 \cdot 10^3 \ldots 2 \cdot 10^9$). Thereby the downward drift of γ on the baseline window is a finite-window subleading curvature, not a dependence of the leading exponent on α; the leading exponent is identically 1, and there is no polynomial α_c. For exponential growth — $\phi_\infty = 1 - e^{-\kappa\tau_E} \in \{0.095;\, 0.394;\, 0.632\} > 0$, $\nu(t) < 1$, but $E_{\text{grow}}(t) \propto e^{\kappa t}$ diverges (ratio up to $9.8 \cdot 10^{42}$ over a run) and $\eta_L(t) \to 0$ through the denominator

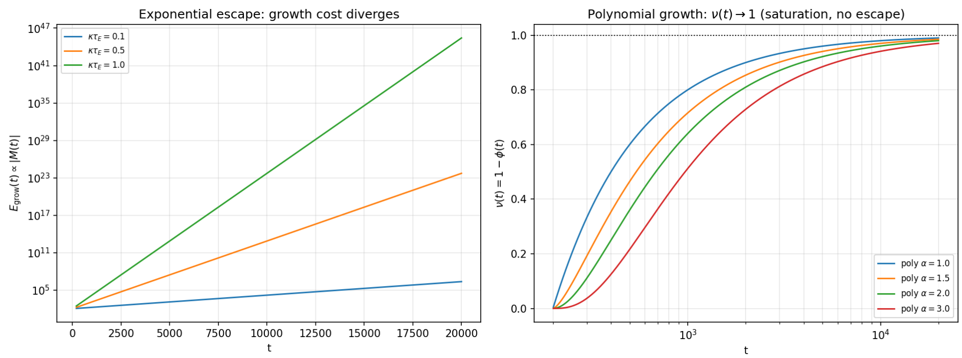

Figure S6. Series `superlinear_memory`, exponential escape and its cost. With $|M_{\le t}| = M_0 e^{\kappa t}$ the fresh fraction reaches a nonzero limit $\varphi_\infty = 1 - e^{-\kappa\tau_E} \in \{0.095; 0.394; 0.632\}$ (escape: $\nu < 1$), but the cumulative cost of capacity expansion $E_{\rm grow}(t) \propto e^{\kappa t}$ diverges (ratio up to $9.8 \cdot 10^{42}$), and the efficiency $\eta_L(t) \to 0$ through the denominator. Escape formally keeps nostalgia below 1, but is thermodynamically ruinous.

(S3.5–S3.6). The saturation $\phi(t) \to 0$ under polynomial growth (the absence of α_c) is shown in Fig. 3 of the main text.

4.5. S4.5 General conventions and limitations

In all series the true $\theta(t)$ is known by construction, so $\nu^{\rm theor} = \nu^{\rm op}$ [1] § 6; $\nu(t)$ without a superscript is used. *Estimator caveat.* In the numerical series of #3 (`fou_floor`, `psp_floor`) nostalgia $\nu(t)$ is read not by the KSG mutual-information estimator, but from a *threshold DPI model of the horizon*: a bit of age Δ is deemed useful if its predictive value by the realised autocorrelation $\hat\rho^2(\Delta)$ (survival on a known $\theta(t)$) exceeds the DPI threshold $\rho^2(\tau_E)$, and nostalgic otherwise. Since $\theta(t)$ is known by construction, conditioning on the latent is realised by *fixing* the true θ, and the conditional $I_{\rm pred} = I(X_t; X_{t+\tau} \mid \theta(t))$ is measured directly — whence $\nu^{\rm theor} = \nu^{\rm op}$ by construction; a separate CMI estimator is not required. The mention of KSG [1] § S6 is inherited from the companion work #2 and *does not apply* to the numerical series of #3 (there θ is known by construction). The simulations illustrate but do not prove the theorems (§ S1–S3 carry the proofs); they do not go beyond abstract Markov chains; they do not cover the non-mixing (heavy-tailed) class outside the domain of Theorem 2. The full implementation, seeds, pseudocode and figure regeneration — in `simulations/{ou_adiab,fou_floor,psp_floor,superlinear_memory}/README.md` of the mirror repository `andriishin/landauer-undertow-oa`.

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